

# SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

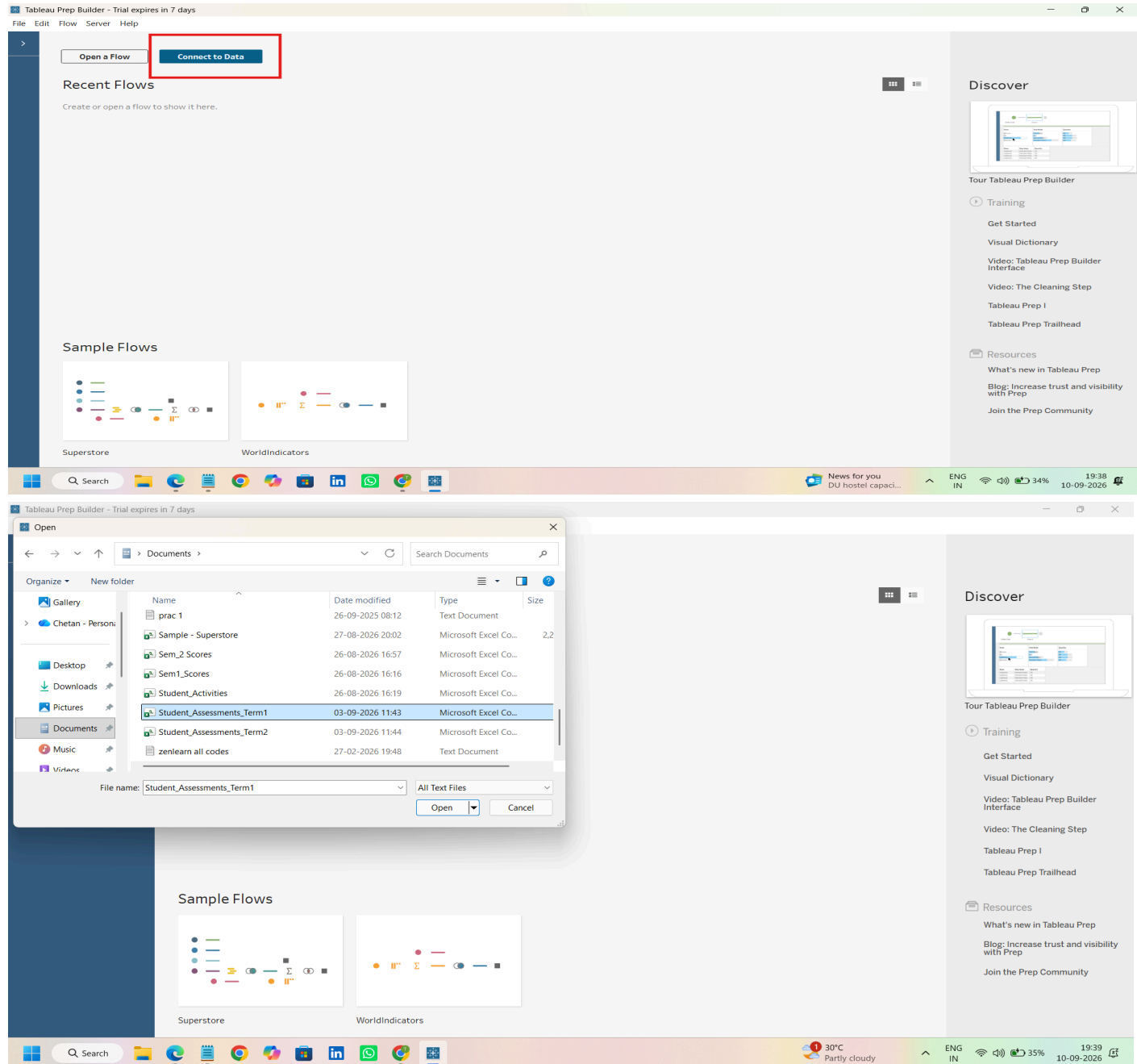
## SUBJECT NAME: Power BI And Tableau Practical Practical 10

Aim:- Data Preparation and Transformation using Tableau

### Steps :-

A: Connect and Load Data into Tableau Prep

Step 1:- Launch Tableau Prep Builder And Click The Connect To Data Icon And Select The CSV File Student\_Assessments\_Term1.csv



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Step 2:- In the bottom Input configuration pane, verify: o Field Separator: Comma (,)  
First row contains header: Checked

The screenshot shows the Tableau Prep Builder interface. In the 'Input' pane, the 'Field Separator' is set to 'Comma' and 'First row contains header' is checked. The 'Data Preview' pane shows a table with the following data:

| Source-Row Nt... | Student_ID | Student_Name | Department  | Assessment | Enrollment_St |
|------------------|------------|--------------|-------------|------------|---------------|
| 1                | S101       | Aarav Sharma | Computer    | Final      | CS101         |
| 2                | S101       | Aarav Sharma | Comp        | Final      | CS101         |
| 3                | S102       | priya VERMA  | Electronics | Midterm    | CS101         |
| 4                | S102       | priya VERMA  | ECE         | Final      | CS101         |
| 5                | S103       | Rohan Gupta  | CS          | Midterm    | CS101         |
| 6                | S103       | Rohan Gupta  | CSE         | Final      | CS101         |
| 7                | S104       | Sneha Patel  | Information | Final      | CS101         |
| 8                | S104       | Sneha Patel  | IT          | Final      | CS101         |
| 9                | S105       | aditya Rao   | null        | null       | null          |

Step 3:- In the top-left Connections pane, click the + (Add connection) button again. Select Text file, browse and select Student\_Assessments\_Term2.csv, and click Open. Click + (Add connection) once more select Text file choose Course\_Credit\_Master.csv click Open.

The screenshot shows the 'Open' dialog box in Tableau Prep Builder. The file 'Student\_Assessments\_Term2' is selected. The 'Data Preview' pane shows the same table as in the previous screenshot.

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Tableau Prep Builder - Flow1\* - Trial expires in 7 days

File Edit Flow Server Help

Connections

- Student\_Assessment... Text file
- Student\_Assessment... Text file

Search

Tables

- Student\_Assessment...

Input 4 fields

Select All Filter Values...

Settings

Header row 1

Data start row 2

Text Options

Field Separator Space

Text Qualifier Automatic

Character Set

Preview Student\_Assessments\_Term2 4 fields

|    |                  | Abc                      | Abc                     | Abc  |
|----|------------------|--------------------------|-------------------------|------|
| 1  | Source Row Nu... | F1                       | F2                      | F3   |
| 2  | ...              | Std_ID,Full_Name,De...   | null                    | null |
| 3  | ...              | core>Total_Possible,E... | null                    | null |
| 4  | ...              | S101,Aarav               | Sharma,Computer         | null |
| 5  | ...              | Science,Midterm,2026...  | null                    | null |
| 6  | ...              | S101,Aarav               | Sharma,CS,Final,2026... | null |
| 7  | ...              | S102,Priya               | Verma,ECE,Midterm,2...  | null |
| 8  | ...              | S102,Priya               | null                    | null |
| 9  | ...              | Verma,Electronics,Fin... | null                    | null |
| 10 | ...              | S103,Rohan               | Gupta,CSE,Midterm,2...  | null |

30°C Partly cloudy

ENG IN

44%

19:47 10-09-2026

Tableau Prep Builder - Flow1\* - Trial expires in 7 days

File Edit Flow Server Help

Connections

- Student\_Assessment... Text file
- Student\_Assessment... Text file
- Course\_Credit\_Mast... Text file

Search

Tables

- Course\_Credit\_Master

Input 5 fields

Select All Filter Values...

Settings

Header row 1

Data start row 2

Text Options

Field Separator Comma

Text Qualifier Automatic

Character Set

Preview Course\_Credit\_Master 5 fields

|   | #                | Abc          | Abc               | #            | #             |
|---|------------------|--------------|-------------------|--------------|---------------|
| 1 | Source Row Nu... | Subject_Code | Subject_Title     | Credit_Hours | Passing_Marks |
| 2 | ...              | CS101        | Data Structures   | 4            | 40            |
| 3 | ...              | CS201        | Algorithms        | 4            | 40            |
| 4 | ...              | EC101        | Digital Logic     | 3            | 40            |
| 5 | ...              | EC201        | Signals & Systems | 4            | 40            |
| 6 | ...              | IT101        | Database Systems  | 4            | 40            |
| 7 | ...              | IT201        | Cloud Computing   | 4            | 40            |
| 8 | ...              | ME101        | Thermodynamics    | 3            | 40            |

30°C Partly cloudy

ENG IN

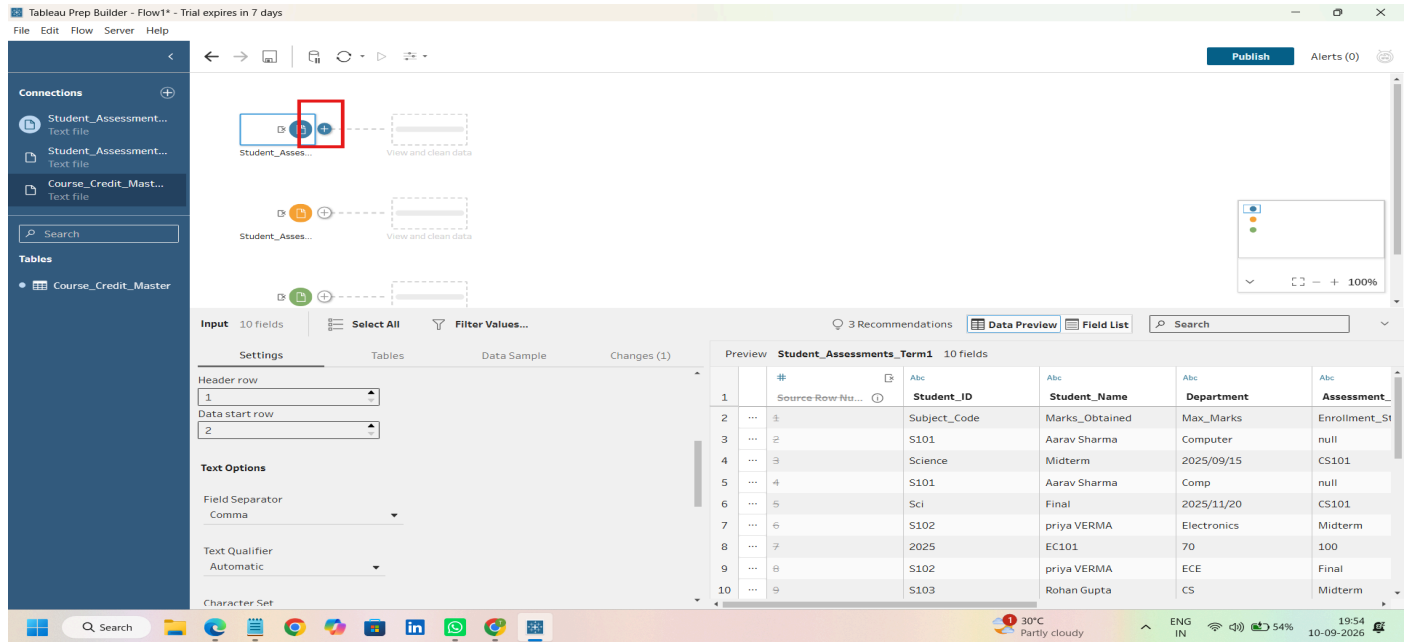
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19:47 10-09-2026

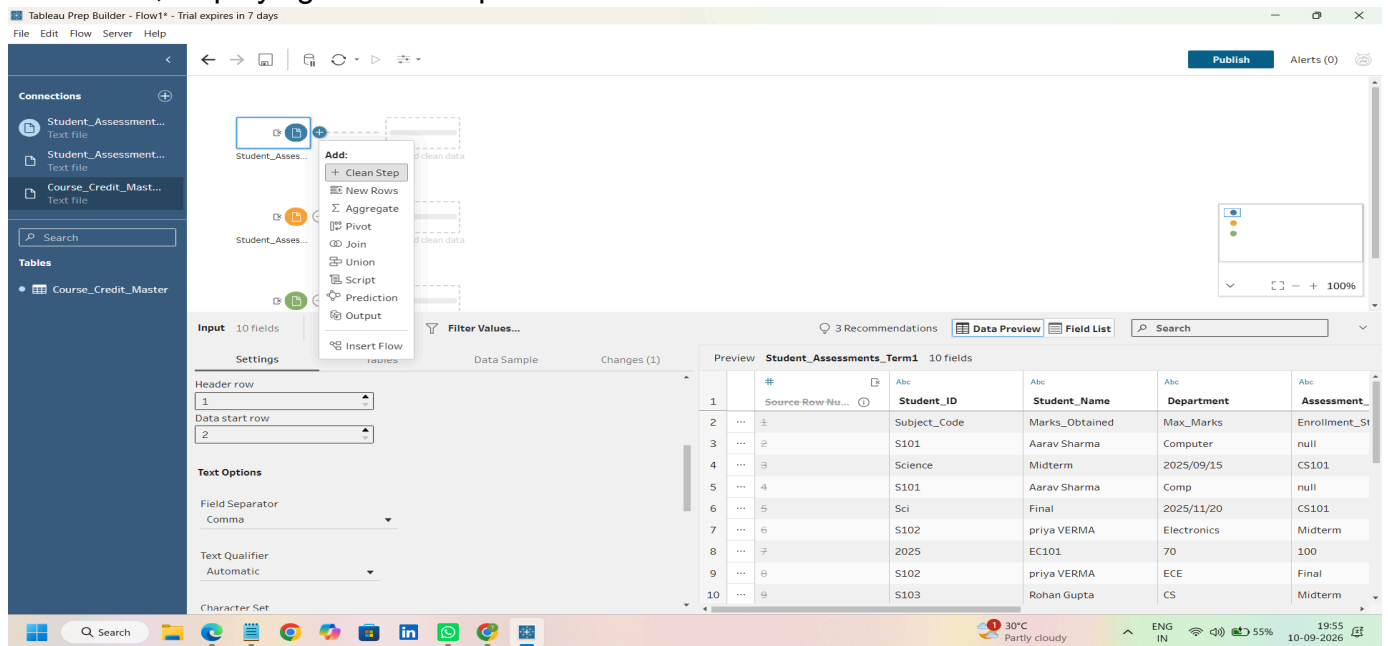
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## SUBJECT NAME: Power BI And Tableau Practical

Step 4:- On the Flow canvas, hover your mouse over the Student\_Assessments\_Term1 input icon. 2. Click the small + icon that appears to its right.



Select Clean Step (the step with the cleaning tool icon). The Profile Pane will open at the bottom, displaying interactive profile cards for each of the 9 columns in Term 1.



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Tableau Prep Builder - Flow1\* - Trial expires in 7 days

File Edit Flow Server Help

Connections

- Student\_Assessment... Text file
- Student\_Assessment... Text file
- Course\_Credit\_Mast... Text file

Search

Tables

- Course\_Credit\_Master

Clean 1 9 fields 26 rows

Select All Filter Values... Identify Duplicate Rows ... 2 Recommendations Search

Student\_ID 15 Student\_Name 13 Department 16 Assessment... 10 Exam\_Date 16 F6 6 F7 11

Changes(0)

Student\_ID 15

Student\_Name 13

Department 16

Assessment... 10

Exam\_Date 16

F6 6

F7 11

29°C Partly cloudy 56% 19:56 10-09-2026

### B: Clean and Profile Data

Step 1:- Click on the Clean 1 step connected to Student\_Assessments\_Term1.

2. Look at the cards displayed in the Profile Pane:

o Each card represents one column.

o Notice that Student\_Name has multiple representations of the same student due to extra spaces and mixed casing (e.g., 'Aarav Sharma' vs 'Aarav Sharma', 'priya VERMA', 'aditya Rao').

Tableau Prep Builder - Flow1\* - Trial expires in 7 days

File Edit Flow Server Help

Connections

- Student\_Assessment... Text file
- Student\_Assessment... Text file
- Course\_Credit\_Mast... Text file

Search

Tables

- Course\_Credit\_Master

Clean 1 9 fields 26 rows

Select All Filter Values... Identify Duplicate Rows ... 2 Recommendations Search

Student\_ID 15 Student\_Name 13 Department 16 Assessment... 10 Exam\_Date 16 F6 6 F7 11

Changes(0)

Student\_ID 15

Student\_Name 13

Department 16

Assessment... 10

Exam\_Date 16

F6 6

F7 11

29°C Partly cloudy 68% 20:08 10-09-2026

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Step 2:- On the Student\_Name profile card: o Click the More options (...) icon at the top right of the card. o Go to Clean --> select Trim Spaces (this strips all leading and trailing whitespace).

Tableau Prep Builder - Flow1\* - Trial expires in 7 days

File Edit Flow Server Help

Connections

- Student\_Assessment... Text file
- Student\_Assessment... Text file
- Course\_Credit\_Mast... Text file

Tables

- Course\_Credit\_Master

Changes (0)

Clean 1 9 fields 26 rows

Select All Filter Values...

Filter

- Clean
- Group Values
- Split Values
- Identify Duplicate Rows
- View State
- Detail
- Summary
- Rename Field
- Duplicate Field
- Keep Only Field
- Create Calculated Field
- Hide Field
- Remove

Make Uppercase

Make Lowercase

Make Titlecase

Remove Letters

Remove Numbers

Remove Punctuation

Trim Spaces

Remove Extra Spaces

Remove All Spaces

| Student_ID | Student_Name   | Department  | Assessment_Score  | Exam_Date  | F6      | F7     |
|------------|----------------|-------------|-------------------|------------|---------|--------|
| 2025       | Aarav Sharma   | 2025/09/15  | 100               | 15-09      | 100     | 100    |
| Engg       | aditya Rao     | 2025/11/20  | 2025/09/15        | 20-11-2025 | 35      | 45     |
| S101       | Ananya Singh   | 70          | CS101             | 2025.09.15 | CS101   | 52     |
| S102       | EC101          | Comp        | EC101             | 2025.11.20 | EC101   | 65     |
| S103       | Final          | Computer    | Enrollment_Status | 2025/09/15 | IT101   | 78     |
| S104       | Kavya Reddy    | CS          | Final             | 2025/11/20 | IT101   | 92     |
| S105       | Marks_Obtained | CSE         | IT101             | 40         | ME101   | 94     |
| S106       | Mechanical     | ECE         | ME101             | 72         | Midterm | 96     |
| S107       | Midterm        | Electronics | Midterm           | 85         | Active  | Droppe |
| S108       | priya VERMA    | Info        |                   | 88         |         |        |
| Sci        | Rahul Nair     | Information |                   | 90         |         |        |
|            | Rohan Gupta    |             |                   |            |         |        |

Click More options (...) again --> Clean --> select Make Uppercase (or Make Lowercase / standard proper formatting).

Tableau Prep Builder - Flow1\* - Trial expires in 7 days

File Edit Flow Server Help

Connections

- Student\_Assessment... Text file
- Student\_Assessment... Text file
- Course\_Credit\_Mast... Text file

Tables

- Course\_Credit\_Master

Changes (1)

Clean 1 9 fields 26 rows

Select All Filter Values...

Filter

- Clean
- Group Values
- Split Values
- Identify Duplicate Rows
- View State
- Detail
- Summary
- Rename Field
- Duplicate Field
- Keep Only Field
- Create Calculated Field
- Hide Field
- Remove

Make Uppercase

Make Lowercase

Make Titlecase

Remove Letters

Remove Numbers

Remove Punctuation

Trim Spaces

Remove Extra Spaces

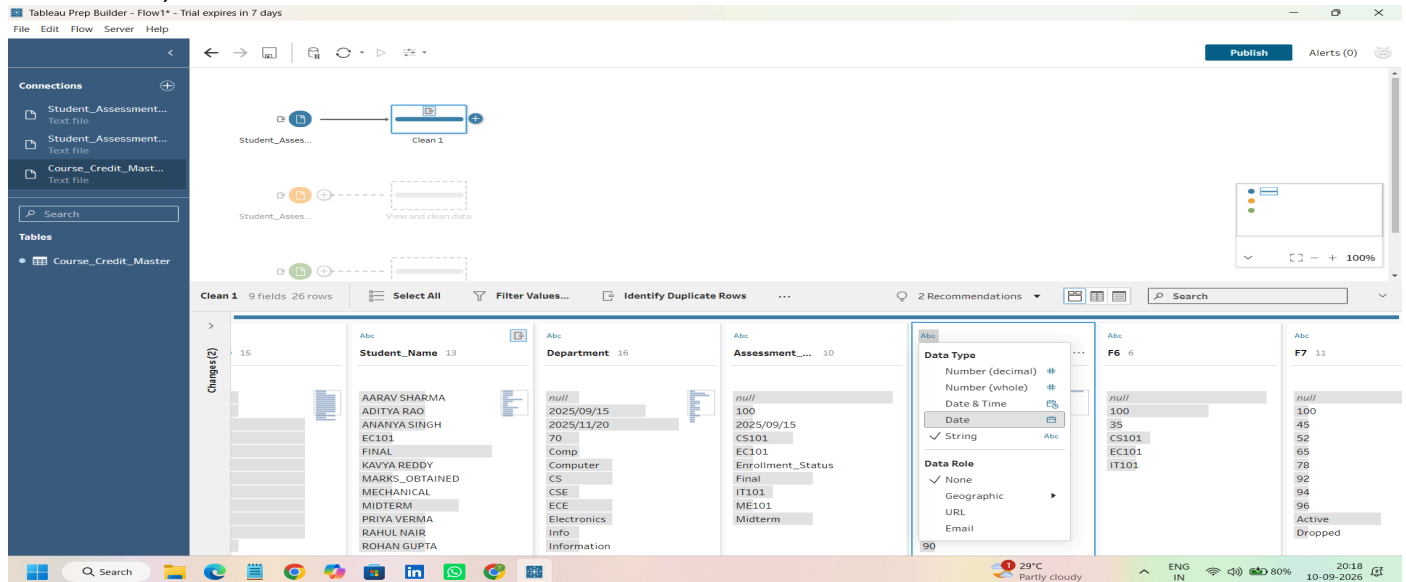
Remove All Spaces

| Student_ID | Student_Name   | Department  | Assessment_Score  | Exam_Date  | F6      | F7     |
|------------|----------------|-------------|-------------------|------------|---------|--------|
| 2025       | Aarav Sharma   | 2025/09/15  | 100               | 15-09      | 100     | 100    |
| Engg       | aditya Rao     | 2025/11/20  | 2025/09/15        | 20-11-2025 | 35      | 45     |
| S101       | Ananya Singh   | 70          | CS101             | 2025.09.15 | CS101   | 52     |
| S102       | EC101          | Comp        | EC101             | 2025.11.20 | EC101   | 65     |
| S103       | Final          | Computer    | Enrollment_Status | 2025/09/15 | IT101   | 78     |
| S104       | Kavya Reddy    | CS          | Final             | 2025/11/20 | IT101   | 92     |
| S105       | Marks_Obtained | CSE         | IT101             | 40         | ME101   | 94     |
| S106       | Mechanical     | ECE         | ME101             | 72         | Midterm | 96     |
| S107       | Midterm        | Electronics | Midterm           | 85         | Active  | Droppe |
| S108       | priya VERMA    | Info        |                   | 88         |         |        |
| Sci        | Rahul Nair     | Information |                   | 90         |         |        |
|            | Rohan Gupta    |             |                   |            |         |        |

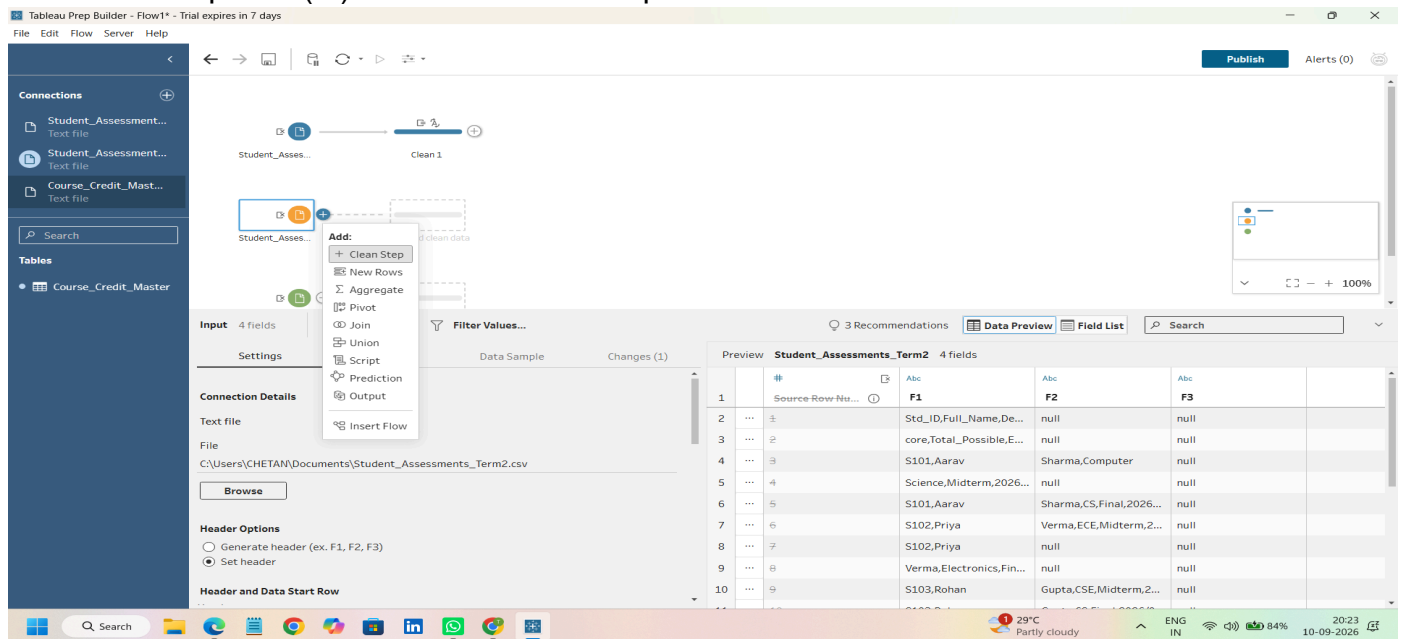
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Step 3:- Locate the Exam\_Date profile card. Click the data type icon (currently showing Abc) directly above the column name. Select Date from the dropdown menu. Tableau Prep automatically parses the various date formats (2025/09/15, 15-09 2025, 2025.11.20) into standardized calendar dates.

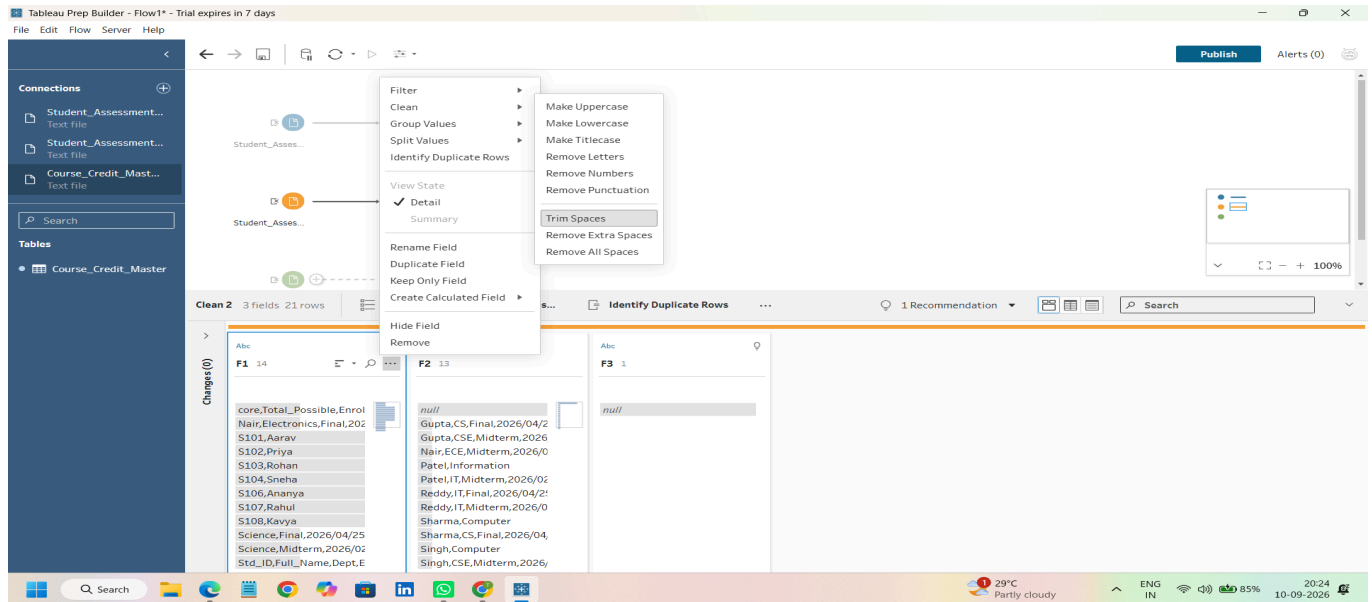


Step 4:- On the Flow canvas, hover over the Student\_Assessments\_Term2 input icon. 2. Click + --> select Clean Step (creates Clean 2). 3. On the Full\_Name card in Clean 2: o Click More options (...) --> Clean --> Trim Spaces.

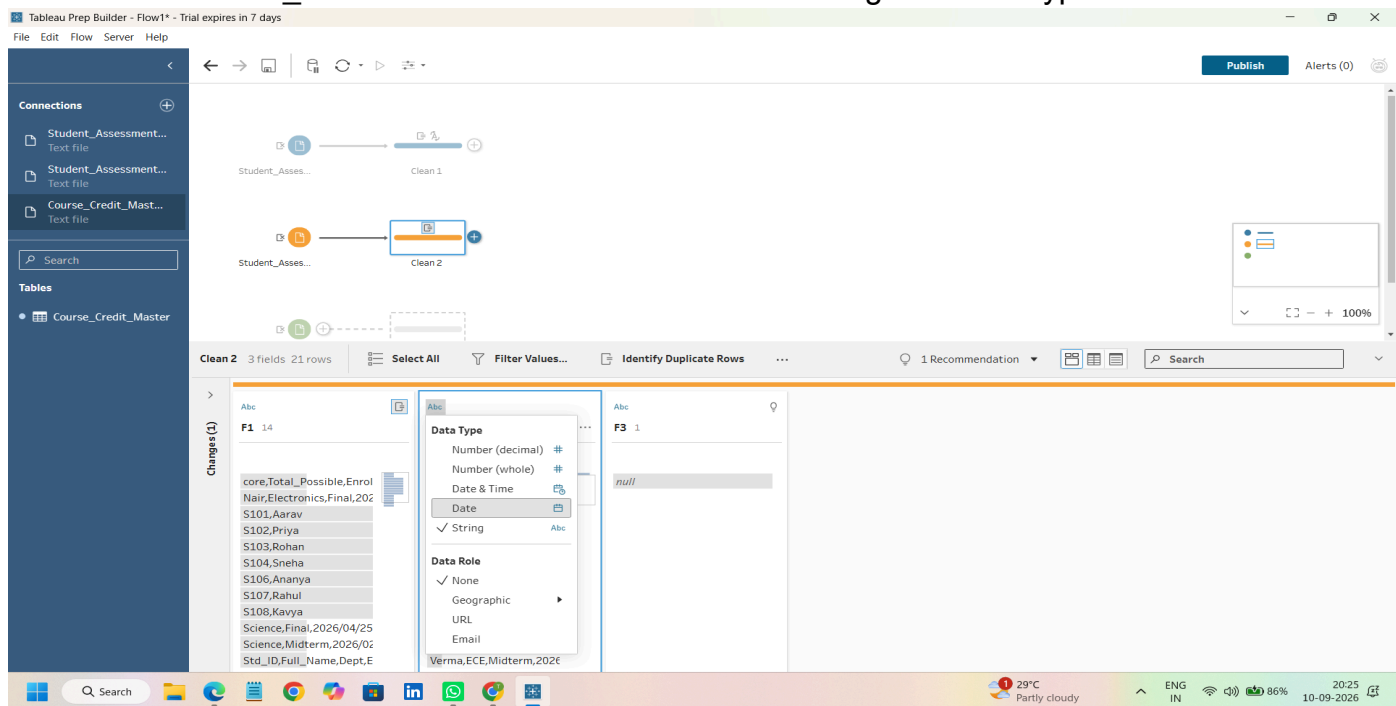


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On the Assessment\_Date card: o Click the Abc icon --> change the data type to Date.



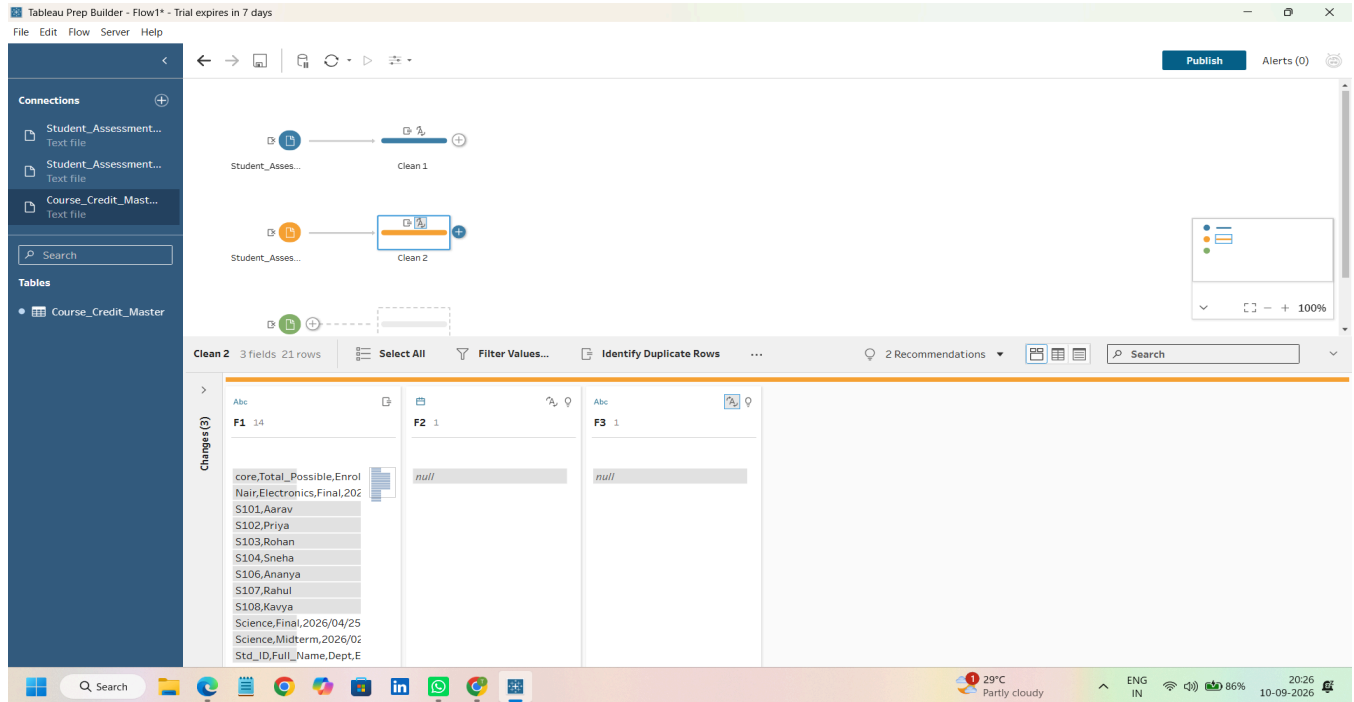
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### C: Group and Standardize Values

Step 1:- Click on the Clean 1 step connected to Student\_Assessments\_Term1.

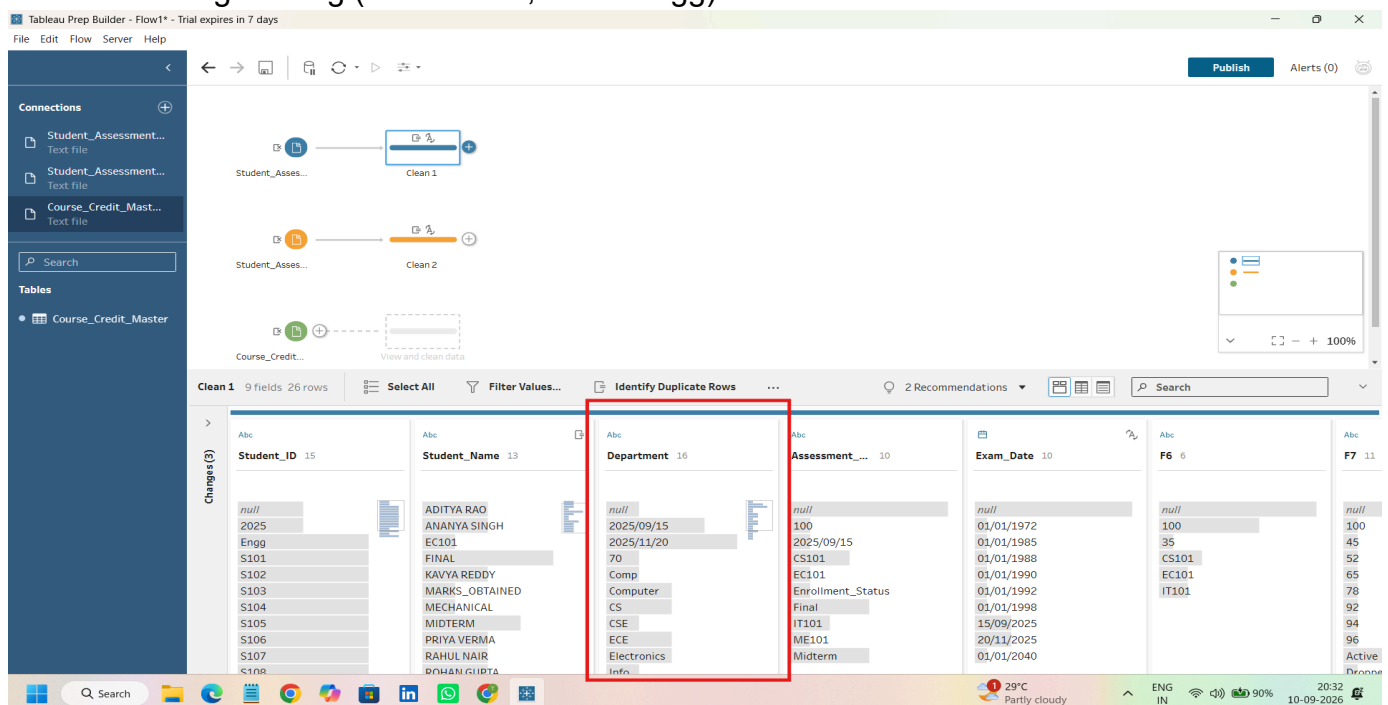
2. Locate the Department card in the Profile Pane.

3. Observe that there are 12 distinct entries representing only 4 actual academic departments: Computer Science (Computer Science, Comp Sci, CS, CSE)

Electronics (Electronics, ECE, Electronics Engg)

Information Technology (Information Tech, IT, Info Tech)

Mechanical Engineering (Mechanical, Mech Engg)



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On the Department card, click the More options (...) menu at the top right. Go to Group Values -->select Manual Selection (or Pronunciation / Common Characters).

The first screenshot shows the Tableau Prep Builder interface with a data flow from 'Student\_Asses...' to 'Clean 1'. The 'Clean 1' card has 9 fields and 26 rows. A context menu is open over the 'Department' field, showing options like 'Filter', 'Clean', 'Group Values', 'Split Values', 'Identify Duplicate Rows', 'View State', 'Detail', 'Summary', 'Rename Field', 'Duplicate Field', 'Keep Only Field', 'Create Calculated Field', 'Hide Field', and 'Remove'. The 'Group Values' option is selected, and a sub-menu is open showing 'Manual Selection', 'Pronunciation', 'Common Characters', and 'Spelling'. The 'Manual Selection' option is chosen.

The second screenshot shows the same interface after the 'Group Values' menu has been opened. The 'Department' card now shows a 'Group Values by Manual Selection' dialog box. The dialog box has a 'Done' button and a message 'No values selected. Select a single value to see the values it replaces.' The 'Department' field is highlighted, and the 'Manual Selection' option is selected in the sub-menu.

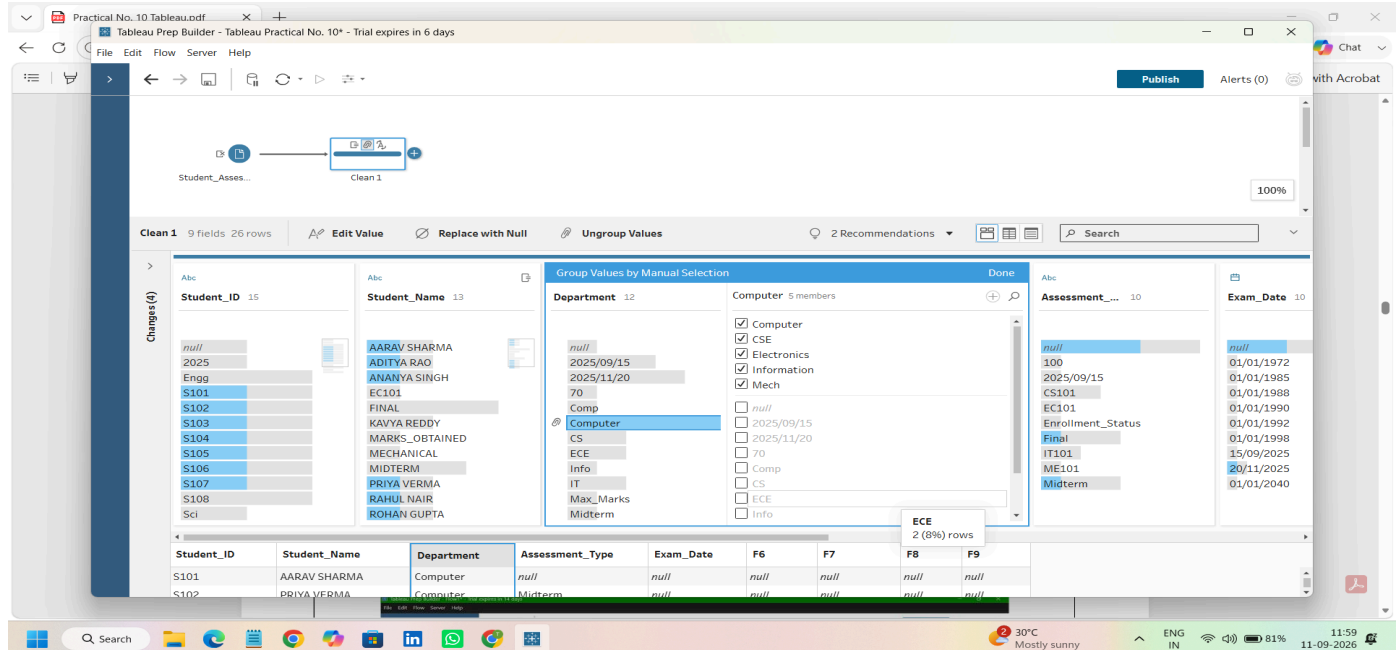
| Student_ID   | Student_Name   | Department | Assessment_Type   | Exam_Date | F6   | F7   | F8   | F9   |
|--------------|----------------|------------|-------------------|-----------|------|------|------|------|
| Subject_Code | MARKS_OBTAINED | Max_Marks  | Enrollment_Status | null      | null | null | null | null |
| S101         | AARAV SHARMA   | Computer   | null              | null      | null | null | null | null |

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Step 2:- In the Grouping editor:

1. Click on Computer Science in the left column. In the right-hand panel, check the boxes for Comp Sci, CS, and CSE
2. Click on Electronics (or rename the master value to Electronics & Communication). Check the boxes for ECE and Electronics Engg.
3. Click on Information Tech (or rename to Information Technology). Check the boxes for IT and Info Tech.
4. Click on Mechanical (or rename to Mechanical Engineering). Check the box for Mech Engg. Click Done at the top right of the editor.



Notice that the paperclip icon appears next to grouped values, consolidating the 12 messy entries into exactly 4 clean departments.

Step 3:- Click on the Clean 2 step connected to Student\_Assessments\_Term2.

2. Locate the Dept card.

3. Click More options (...) -->Group Values -->Manual Selection:

o Group CS, CSE, and Computer Science into Computer Science.

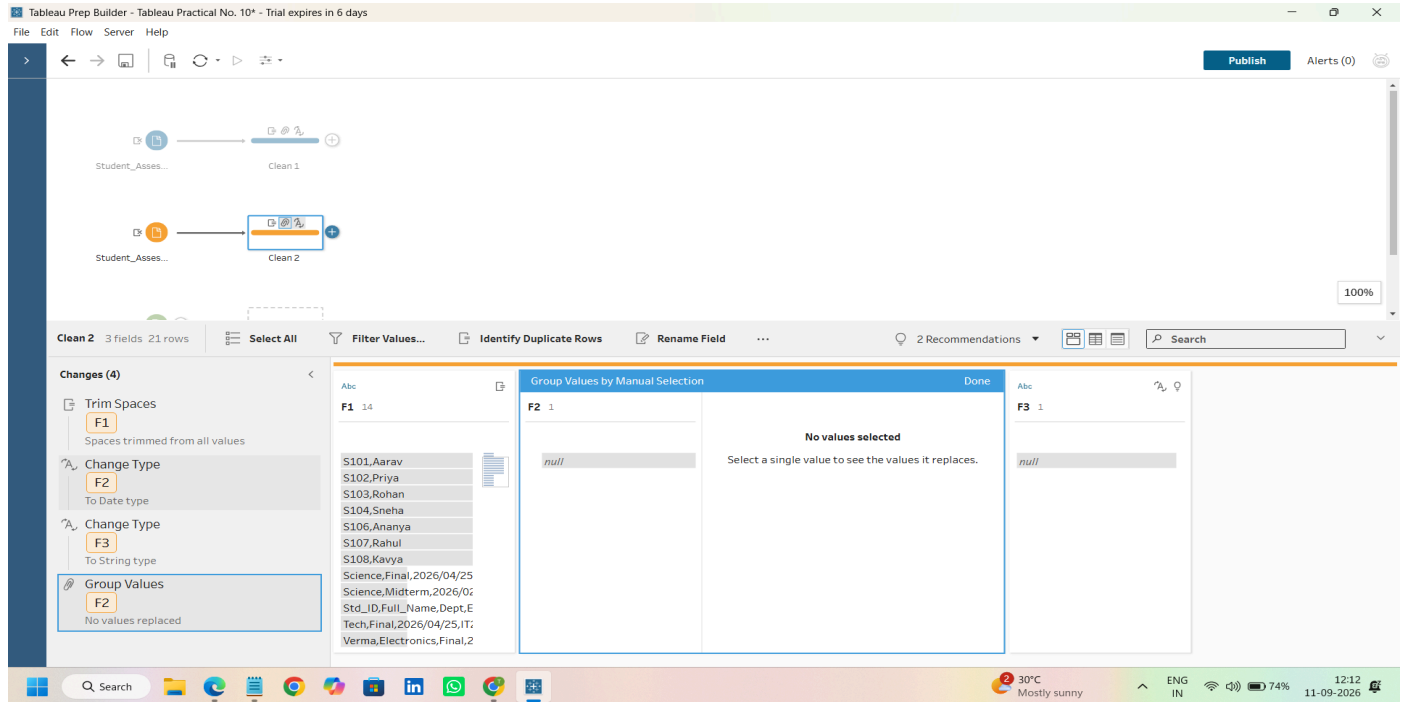
o Group ECE and Electronics into Electronics.

o Group IT and Information Tech into Information Tech.

4. Click Done.

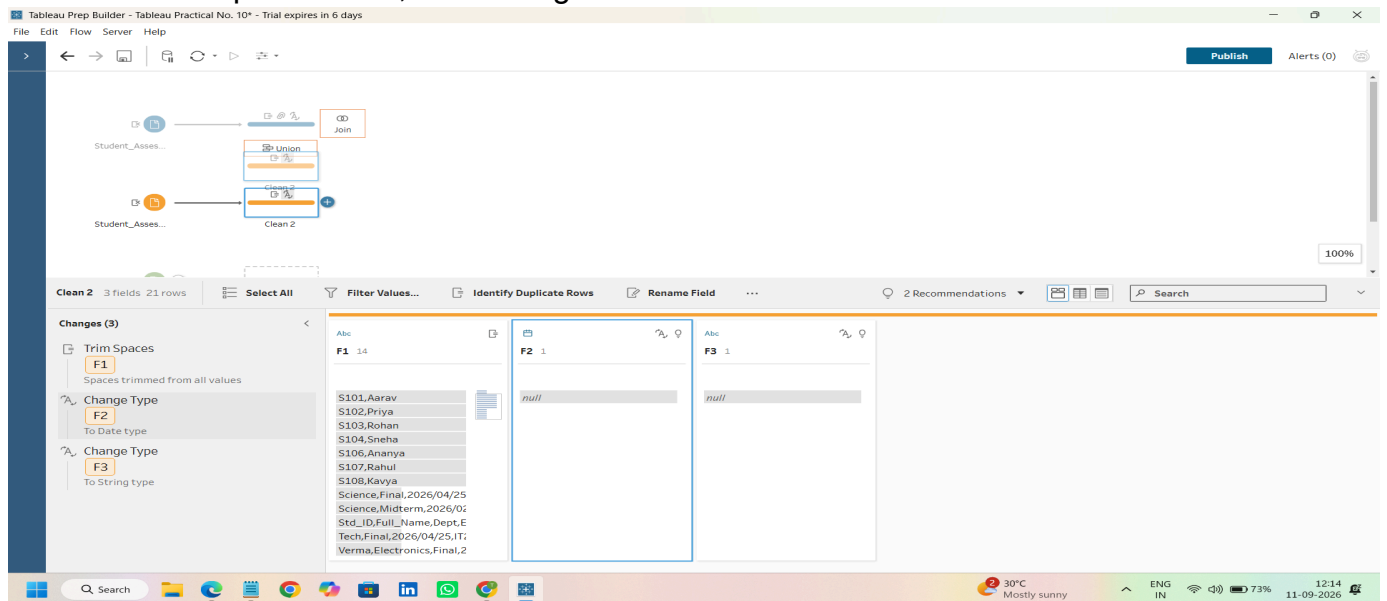
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## SUBJECT NAME: Power BI And Tableau Practical



### D: Merge Mismatched Fields

Step 1:- On the Flow canvas, click and drag the Clean 2 step icon over to the Clean 1 step icon. When the drop zone appears, hover over Union and release the mouse. A new Union 1 step is created, connecting both transformation branches.



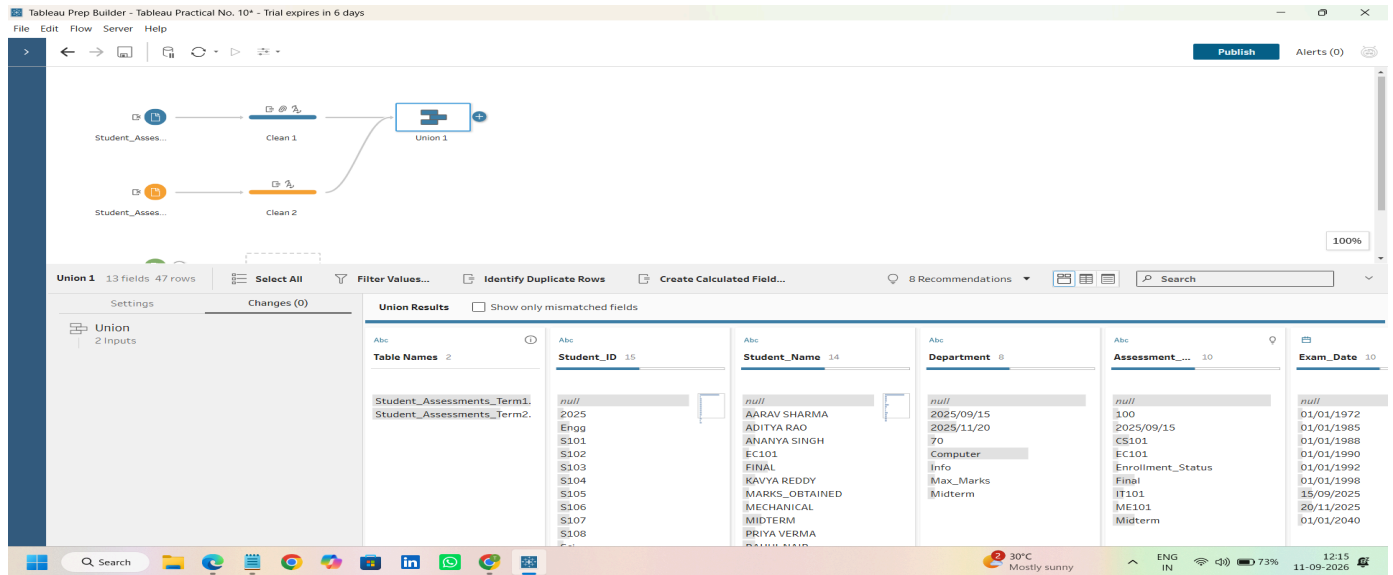
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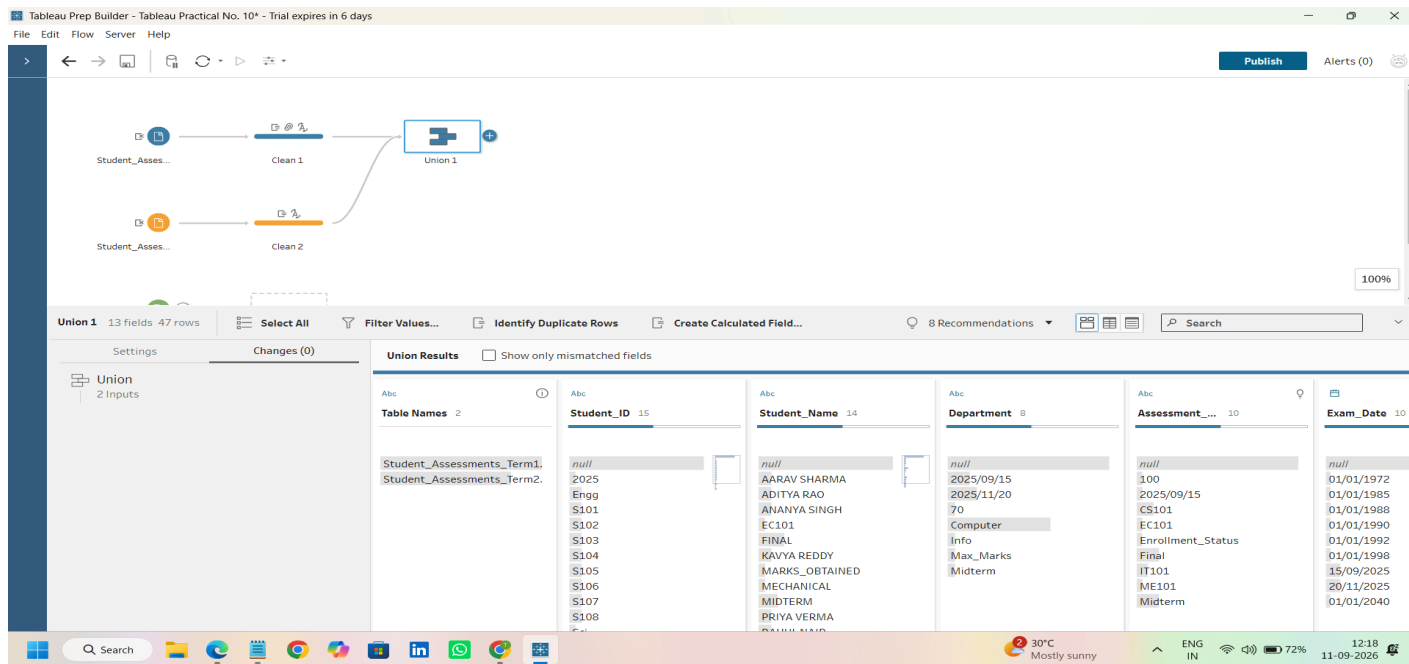
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Step 2:- Click on the Union 1 step. In the bottom Union summary pane, look at the Mismatched Fields section.



Notice that because the two files used different naming conventions, Tableau Prep lists non-matching pairs (indicated by yellow and blue bars), such as:

- o Student\_ID (Term 1) vs Std\_ID (Term 2)
- o Student\_Name (Term 1) vs Full\_Name (Term 2)
- o Department (Term 1) vs Dept (Term 2)
- o Assessment\_Type (Term 1) vs Exam\_Type (Term 2)
- o Exam\_Date (Term 1) vs Assessment\_Date (Term 2)
- o Subject\_Code (Term 1) vs Course\_Code (Term 2)
- o Marks\_Obtained (Term 1) vs Score (Term 2)
- o Max\_Marks (Term 1) vs Total\_Possible (Term 2)

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Tableau Prep Builder - Tableau Practical No. 10\* - Trial expires in 6 days

File Edit Flow Server Help

Union 1 13 fields 47 rows

Settings Changes (0)

Union Results Show only mismatched fields

| Table Names               | Student_ID | Student_Name   | Department | Assessment...     | Exam_Date  |
|---------------------------|------------|----------------|------------|-------------------|------------|
| Student_Assessments_Term1 | 2025       | ABHAY SHARMA   | 2025/09/15 | 100               | 01/01/1972 |
| Student_Assessments_Term2 | Engrg      | ADITYA RAO     | 2025/11/20 | 100               | 01/01/1985 |
|                           | S101       | ANANYA SINGH   | 70         | CS101             | 01/01/1988 |
|                           | S102       | EC101          | Computer   | EC101             | 01/01/1990 |
|                           | S103       | FINAL          | Info       | Enrollment_Status | 01/01/1992 |
|                           | S104       | KAVYA REDDY    | Max_Marks  | Final             | 01/01/1998 |
|                           | S105       | MARKS_OBTAINED | Midterm    | IT101             | 15/09/2025 |
|                           | S106       | MECHANICAL     |            | ME101             | 20/11/2025 |
|                           | S107       | MIDTERM        |            | Midterm           | 01/01/2040 |
|                           | S108       | PRIYA VERMA    |            |                   |            |

Step 3:- In the Mismatched Fields list (or the union profile pane):

- o Click and drag Std\_ID directly on top of Student\_ID to merge them into one column.
- o Drag Full\_Name onto Student\_Name.
- o Drag Dept onto Department.

Tableau Prep Builder - Prep10\* - Trial expires in 14 days

File Edit Flow Server Help

Union 1 22 fields 58 rows

Settings Changes (0)

Union Results Show only mismatched fields

Mismatched Fields

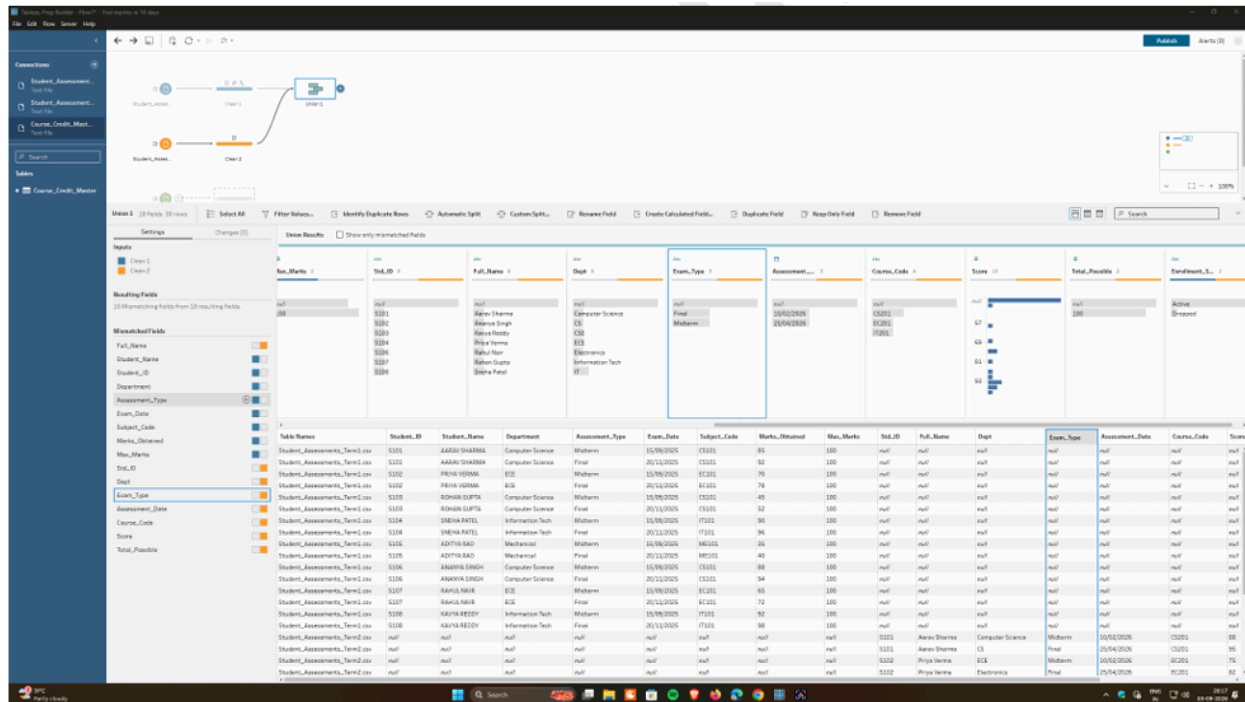
- Std\_ID
- Full\_Name
- Student\_Name
- Student\_ID
- Department
- Assessment\_Type
- Exam\_Date
- Subject\_Code
- Marks\_Obtained
- Max\_Marks
- Std\_ID
- Full\_Name
- Dept
- Exam\_Type
- Assessment\_Date
- Course\_Code
- Score

| Table Names               | Student_ID | Student_Name | Department       | Assessment_Type | Exam_Date  | Subject_Code | Marks_Obtained | Max_Marks | Std_ID | Full_Name    | Dept             | Exam_Type | Assessment_Date | Course_Code | Score |
|---------------------------|------------|--------------|------------------|-----------------|------------|--------------|----------------|-----------|--------|--------------|------------------|-----------|-----------------|-------------|-------|
| Student_Assessments_Term1 | S101       | ABHAY SHARMA | Computer Science | Midterm         | 15/09/2025 | CS101        | 85             | 100       | 2025   | ABHAY SHARMA | Computer Science | Midterm   | 15/09/2025      | CS101       | 85    |
| Student_Assessments_Term2 | S102       | ADITYA RAO   | Computer Science | Final           | 20/11/2025 | CS102        | 92             | 100       | 2025   | ADITYA RAO   | Computer Science | Final     | 20/11/2025      | CS102       | 92    |
| Student_Assessments_Term1 | S103       | ANANYA SINGH | EC               | Final           | 15/09/2025 | EC101        | 78             | 100       | 2025   | ANANYA SINGH | EC               | Final     | 15/09/2025      | EC101       | 78    |
| Student_Assessments_Term2 | S104       | KAVYA REDDY  | Information Tech | Midterm         | 20/11/2025 | IT101        | 75             | 100       | 2025   | KAVYA REDDY  | Information Tech | Midterm   | 20/11/2025      | IT101       | 75    |
| Student_Assessments_Term1 | S105       | MECHANICAL   | Information Tech | Final           | 20/11/2025 | ME101        | 88             | 100       | 2025   | MECHANICAL   | Information Tech | Final     | 20/11/2025      | ME101       | 88    |
| Student_Assessments_Term2 | S106       | ADITYA RAO   | Computer Science | Midterm         | 15/09/2025 | CS102        | 82             | 100       | 2025   | ADITYA RAO   | Computer Science | Midterm   | 15/09/2025      | CS102       | 82    |
| Student_Assessments_Term1 | S107       | ABHAY SHARMA | Computer Science | Final           | 20/11/2025 | CS101        | 95             | 100       | 2025   | ABHAY SHARMA | Computer Science | Final     | 20/11/2025      | CS101       | 95    |
| Student_Assessments_Term2 | S108       | ANANYA SINGH | EC               | Final           | 15/09/2025 | EC101        | 80             | 100       | 2025   | ANANYA SINGH | EC               | Final     | 15/09/2025      | EC101       | 80    |
| Student_Assessments_Term1 | S109       | KAVYA REDDY  | Information Tech | Midterm         | 20/11/2025 | IT101        | 72             | 100       | 2025   | KAVYA REDDY  | Information Tech | Midterm   | 20/11/2025      | IT101       | 72    |
| Student_Assessments_Term2 | S110       | MECHANICAL   | Information Tech | Final           | 20/11/2025 | ME101        | 85             | 100       | 2025   | MECHANICAL   | Information Tech | Final     | 20/11/2025      | ME101       | 85    |
| Student_Assessments_Term1 | S111       | ADITYA RAO   | Computer Science | Midterm         | 15/09/2025 | CS102        | 88             | 100       | 2025   | ADITYA RAO   | Computer Science | Midterm   | 15/09/2025      | CS102       | 88    |
| Student_Assessments_Term2 | S112       | ABHAY SHARMA | Computer Science | Final           | 20/11/2025 | CS101        | 90             | 100       | 2025   | ABHAY SHARMA | Computer Science | Final     | 20/11/2025      | CS101       | 90    |
| Student_Assessments_Term1 | S113       | ANANYA SINGH | EC               | Final           | 15/09/2025 | EC101        | 75             | 100       | 2025   | ANANYA SINGH | EC               | Final     | 15/09/2025      | EC101       | 75    |
| Student_Assessments_Term2 | S114       | KAVYA REDDY  | Information Tech | Midterm         | 20/11/2025 | IT101        | 70             | 100       | 2025   | KAVYA REDDY  | Information Tech | Midterm   | 20/11/2025      | IT101       | 70    |
| Student_Assessments_Term1 | S115       | MECHANICAL   | Information Tech | Final           | 20/11/2025 | ME101        | 80             | 100       | 2025   | MECHANICAL   | Information Tech | Final     | 20/11/2025      | ME101       | 80    |
| Student_Assessments_Term2 | S116       | ADITYA RAO   | Computer Science | Midterm         | 15/09/2025 | CS102        | 85             | 100       | 2025   | ADITYA RAO   | Computer Science | Midterm   | 15/09/2025      | CS102       | 85    |
| Student_Assessments_Term1 | S117       | ABHAY SHARMA | Computer Science | Final           | 20/11/2025 | CS101        | 92             | 100       | 2025   | ABHAY SHARMA | Computer Science | Final     | 20/11/2025      | CS101       | 92    |
| Student_Assessments_Term2 | S118       | ANANYA SINGH | EC               | Final           | 15/09/2025 | EC101        | 78             | 100       | 2025   | ANANYA SINGH | EC               | Final     | 15/09/2025      | EC101       | 78    |
| Student_Assessments_Term1 | S119       | KAVYA REDDY  | Information Tech | Midterm         | 20/11/2025 | IT101        | 75             | 100       | 2025   | KAVYA REDDY  | Information Tech | Midterm   | 20/11/2025      | IT101       | 75    |
| Student_Assessments_Term2 | S120       | MECHANICAL   | Information Tech | Final           | 20/11/2025 | ME101        | 82             | 100       | 2025   | MECHANICAL   | Information Tech | Final     | 20/11/2025      | ME101       | 82    |

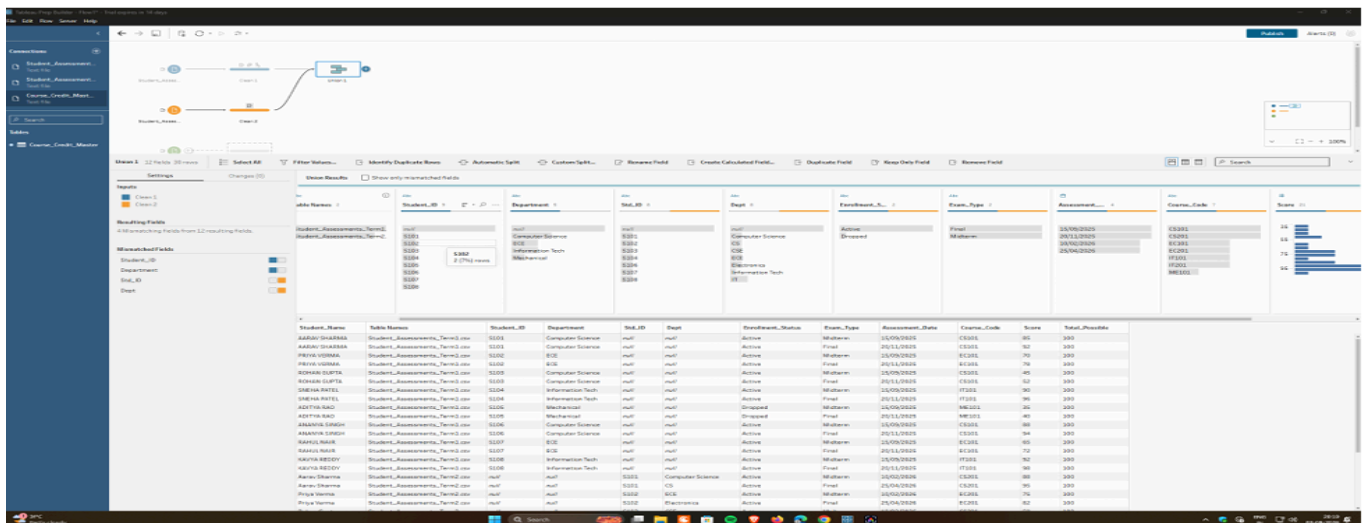
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## SUBJECT NAME: Power BI And Tableau Practical

Drag Exam\_Type onto Assessment\_Type



- o Drag Assessment\_Date onto Exam\_Date.
- o Drag Course\_Code onto Subject\_Code.
- o Drag Score onto Marks\_Obtained.
- o Drag Total\_Possible onto Max\_Marks.



Notice that as you merge them, the mismatched count drops to 0, and each unified column displays both source colors.

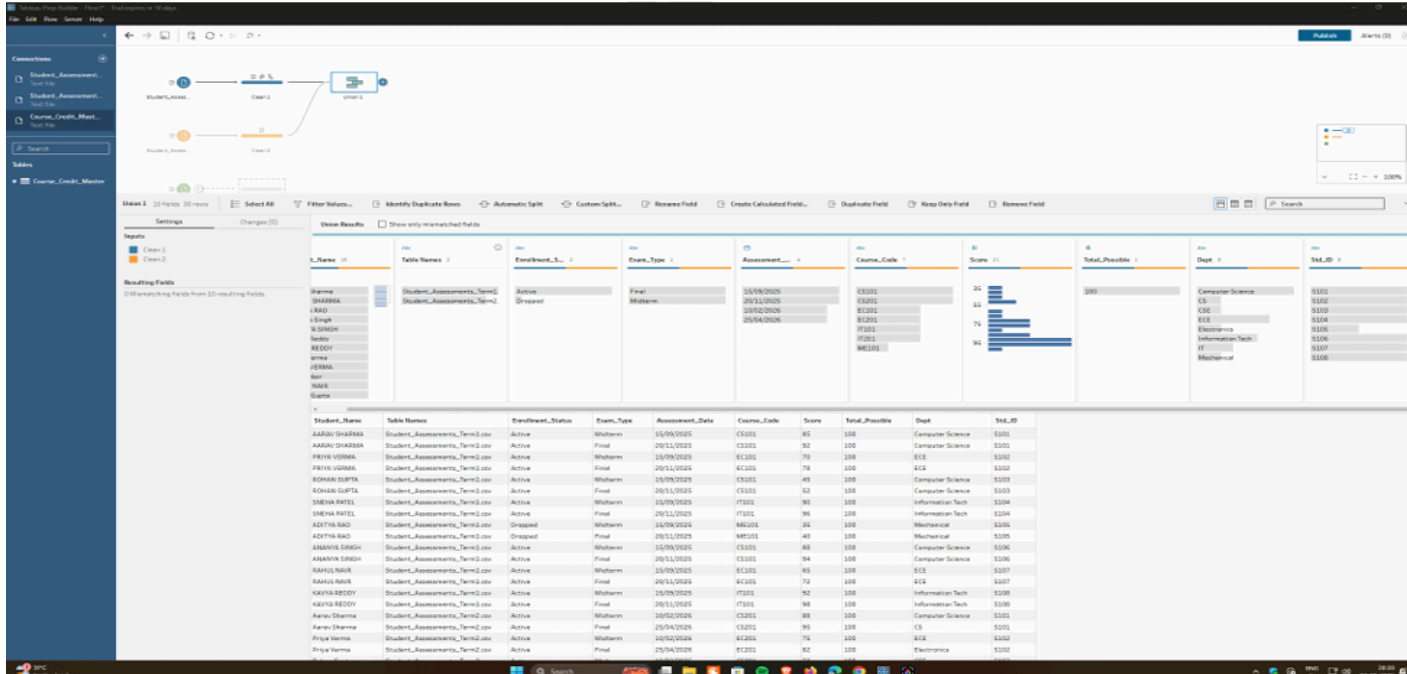
T101

RAJDEEP M PARAB

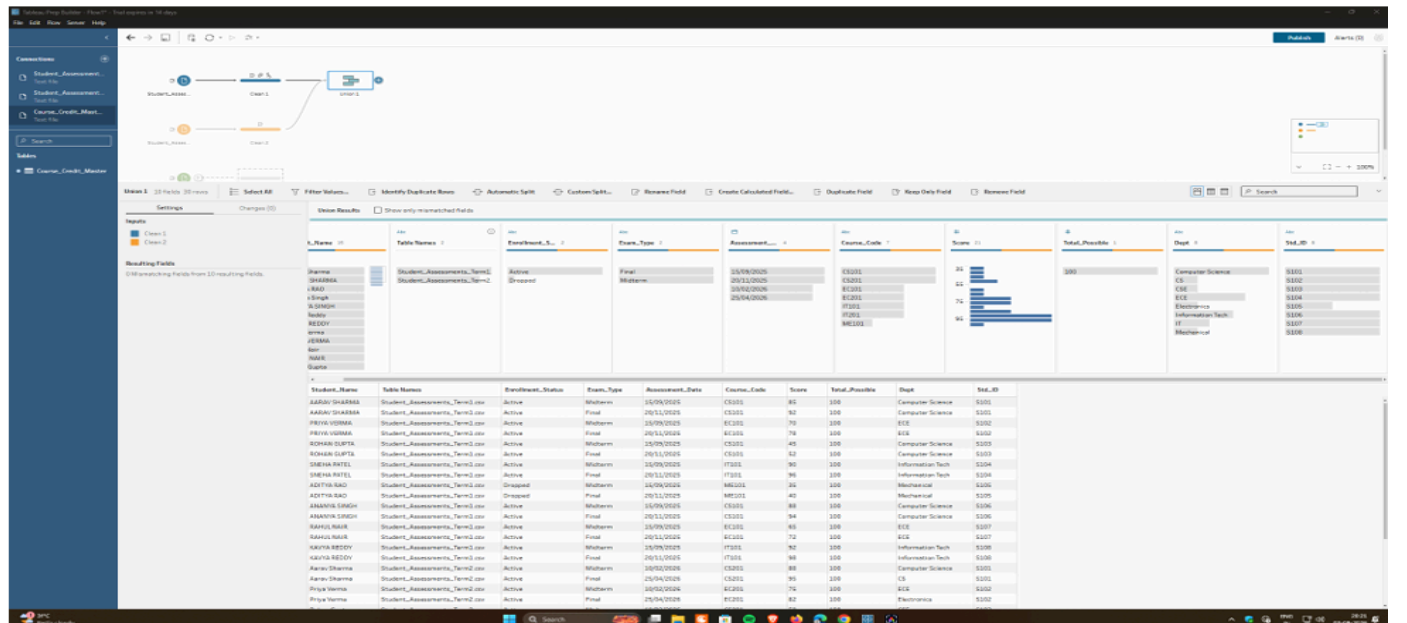
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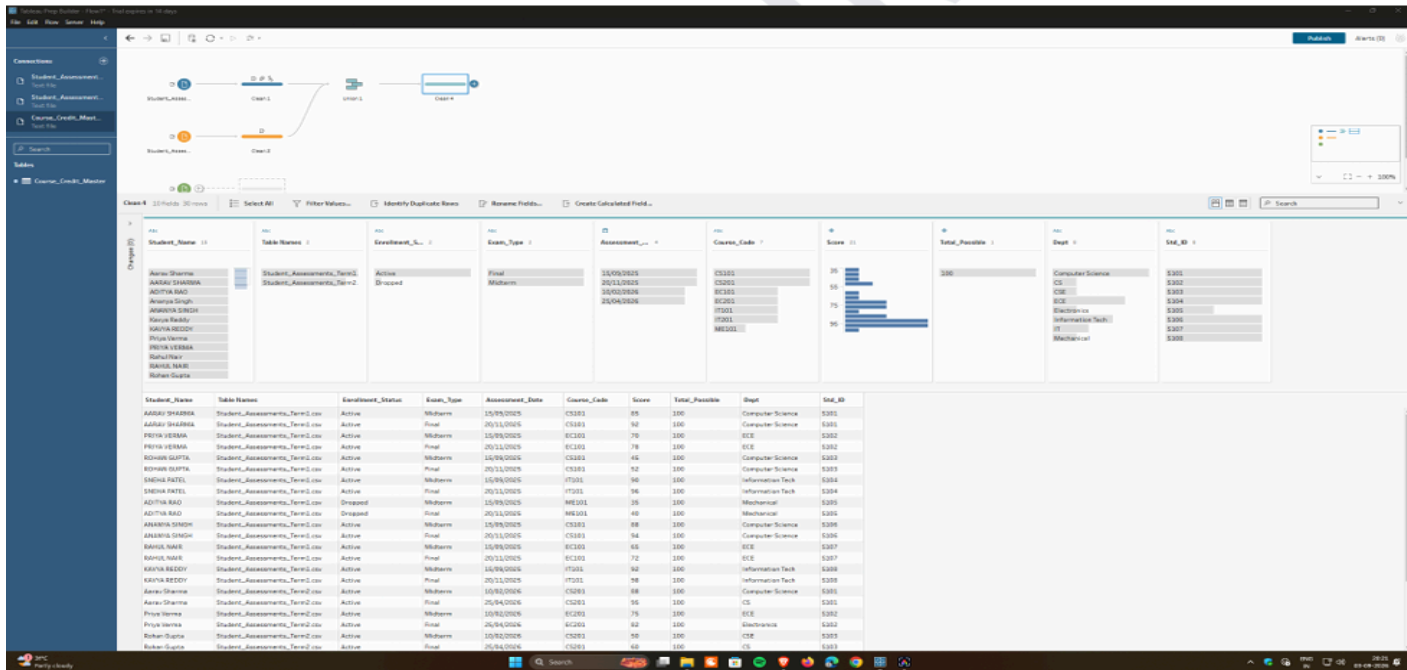
Step 4:- Click the + icon next to the Union 1 node.



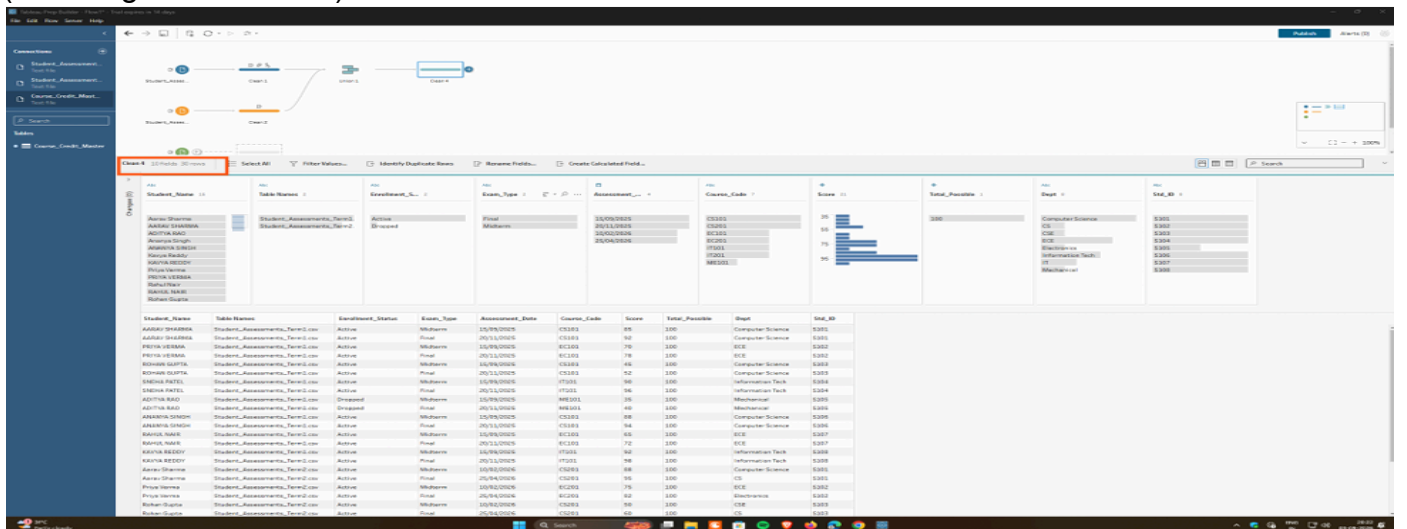
Select Clean Step (creates Clean 3). In Screenshot it is showing Clean 4 because I have created clean step for Course\_Credit\_Master.csv too.

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Verify that the combined dataset now contains 30 total rows across 10 unified columns (including Table Names).

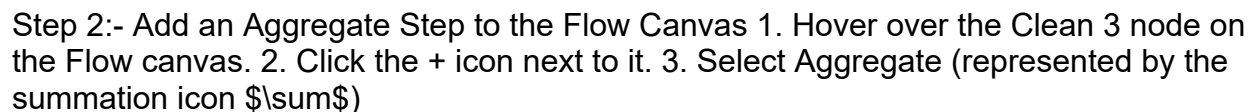


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Step 1:- 1.Click on the Clean 3 step (the clean step created after Union 1).

### 3.Set the Field Name to: Term

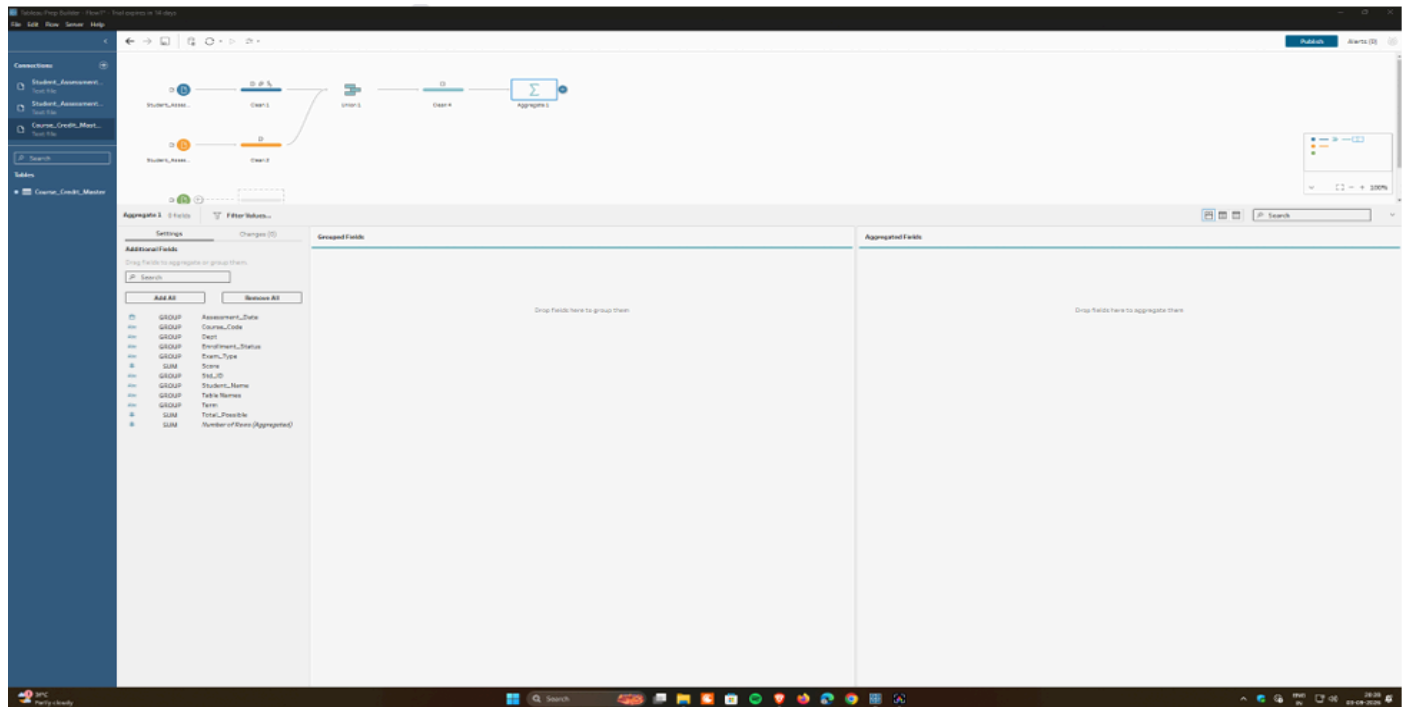
5. Click **Save**.



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4. An Aggregate 1 step will appear, showing two drop zones in the configuration pane: o Grouped Fields (Left Zone) o Aggregated Fields (Right Zone)

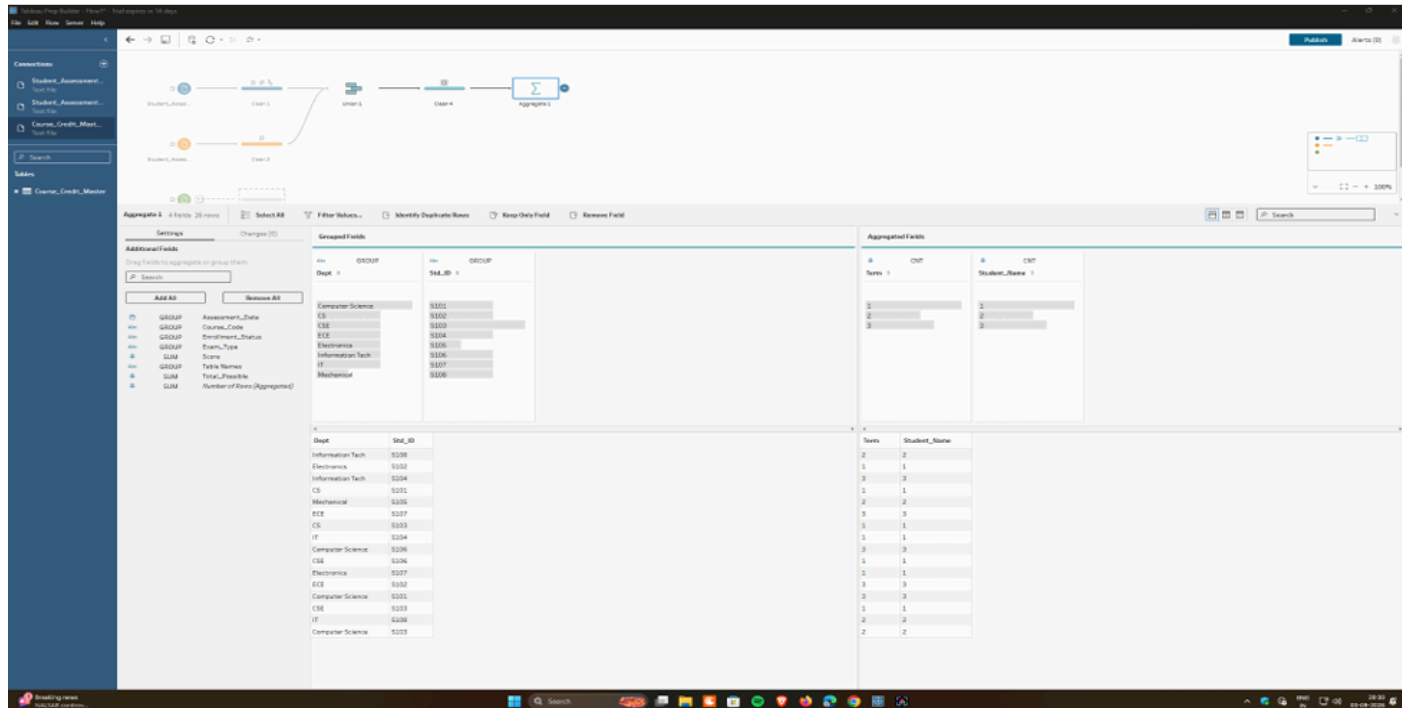


Step 3:- Configure Grouped Fields (Dimensions) From the left-hand column list in the Aggregate step pane Your Column names could be different based on how you did previous steps of Union:

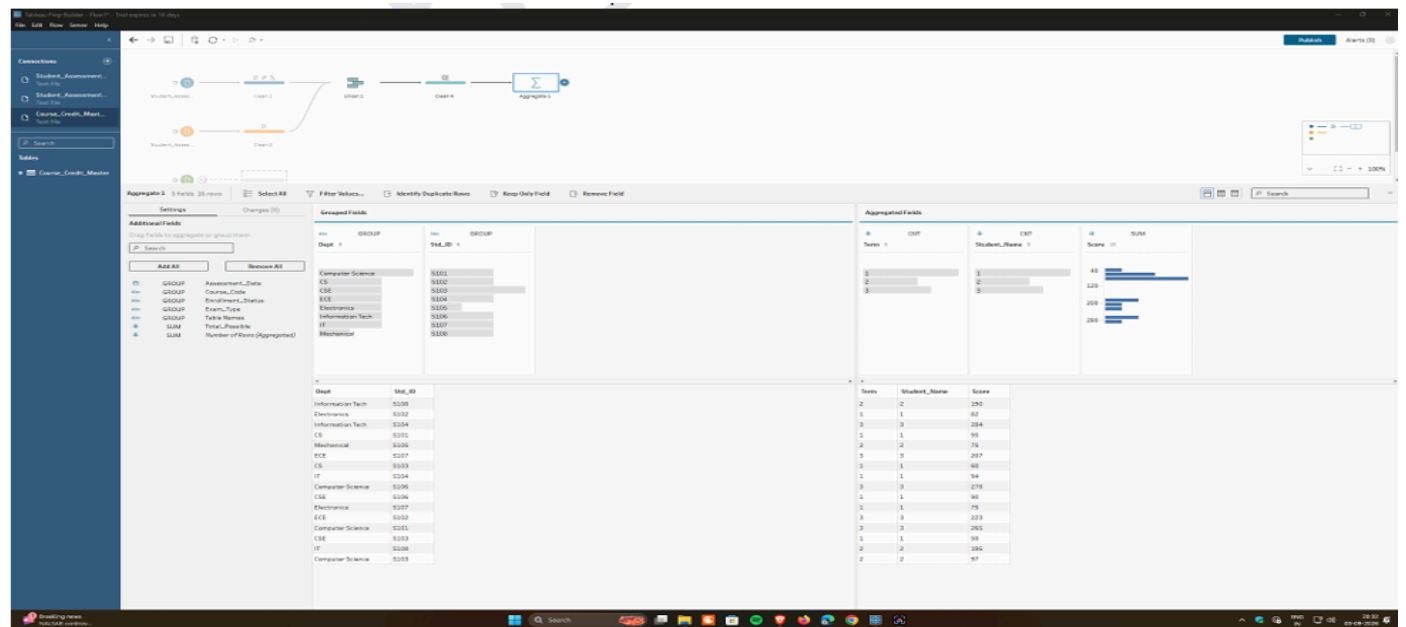
1. Drag Student\_ID into the Grouped Fields box.
2. Drag Student\_Name into the Grouped Fields box.
3. Drag Department into the Grouped Fields box.
4. Drag Term into the Grouped Fields box.

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Step 4: Configure Aggregated Fields (Measures & Metrics) 1. Total Marks: Drag Marks\_Obtained into the Aggregated Fields box. (Ensure aggregation operation is set to Sum).



Average Score: Drag Marks\_Obtained into the Aggregated Fields box a second time --> click on Sum --> change it to Average. There will be no field for dragging by any chance just change SUM to AVERAGE of the existing one.

Total Possible Marks: Drag Max\_Marks into the Aggregated Fields box --> keep as Sum.

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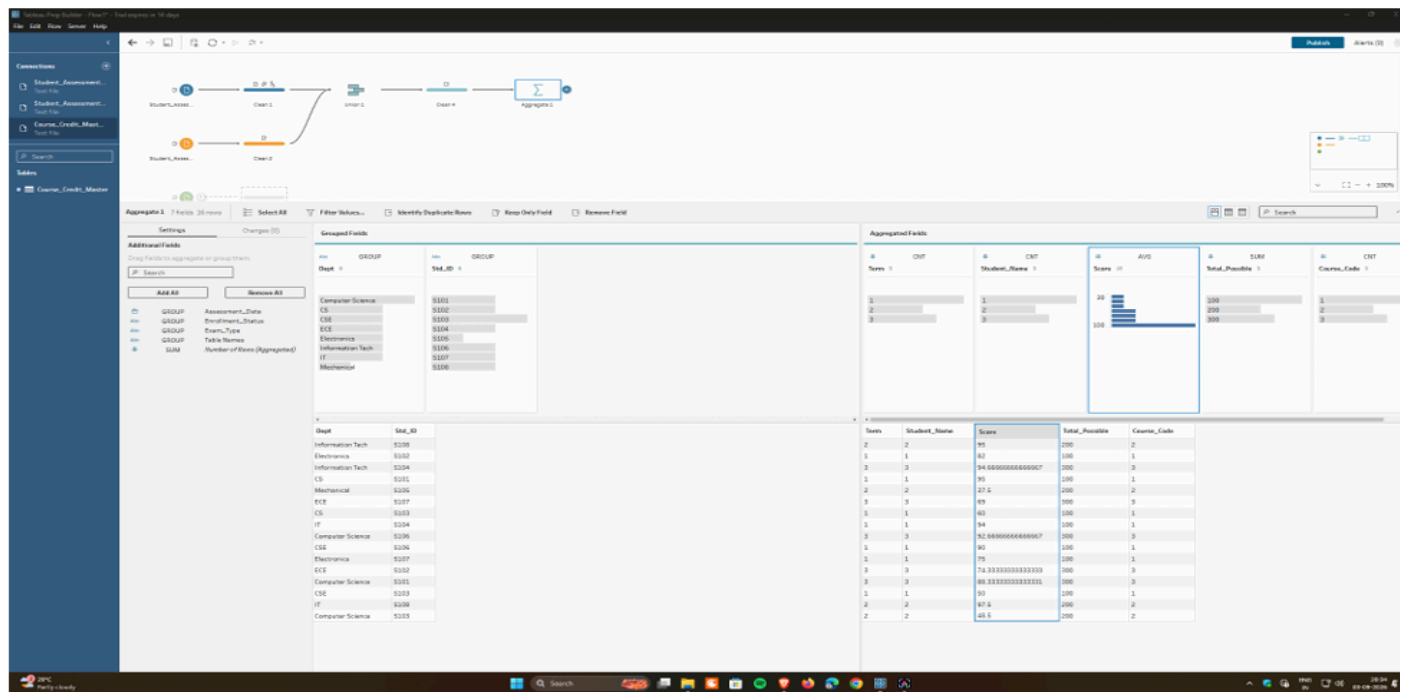
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Assessment Count: Drag Assessment\_Type (or Subject\_Code) into the Aggregated Fields box --> set aggregation to Count (or Count Distinct).



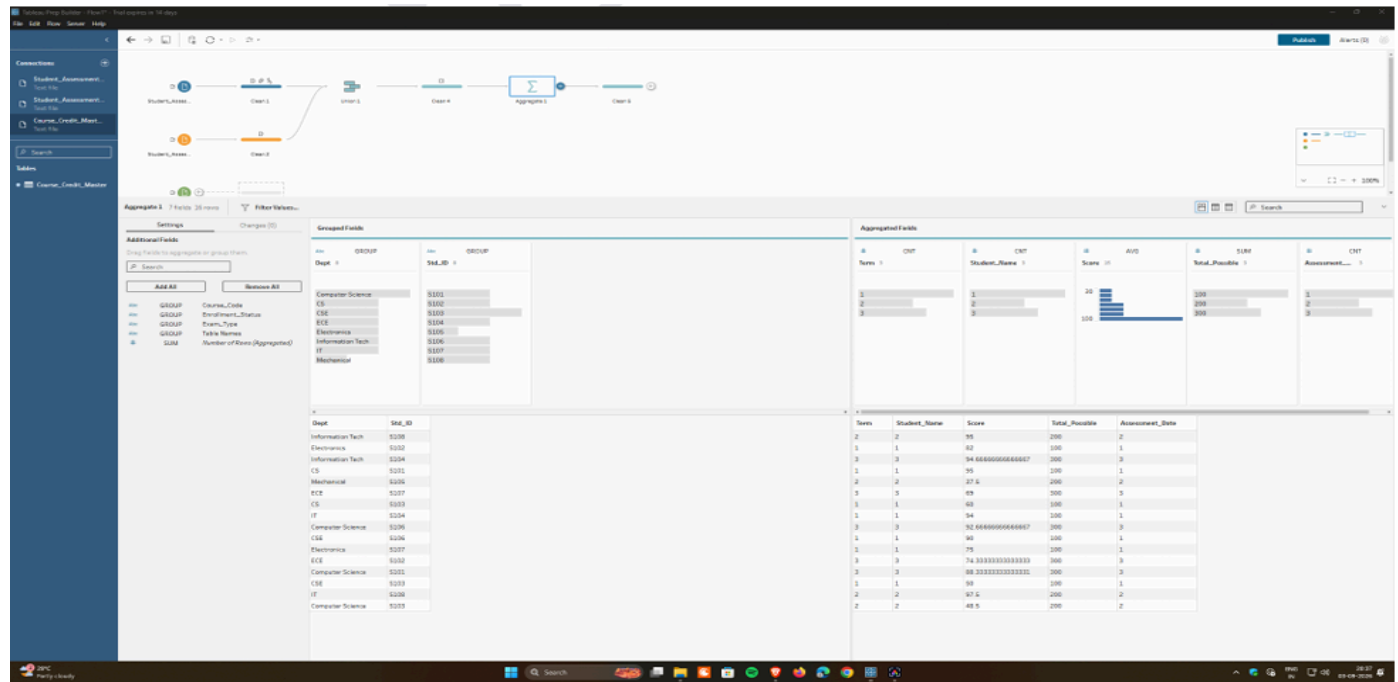
Rename Aggregated Output Fields In the Aggregated Fields panel, click on each field name to rename them cleanly:

- Marks\_Obtained (Sum) --> Total\_Marks\_Obtained
- Marks\_Obtained (Average) --> Average\_Score
- Max\_Marks (Sum) --> Total\_Max\_Marks
- Assessment\_Type (Count) --> Assessments\_Taken

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Step 5:- Add a Clean Step to Inspect Aggregated Output 1. Click the + icon next to the Aggregate 1 node --> select Clean Step (creates Clean 4).



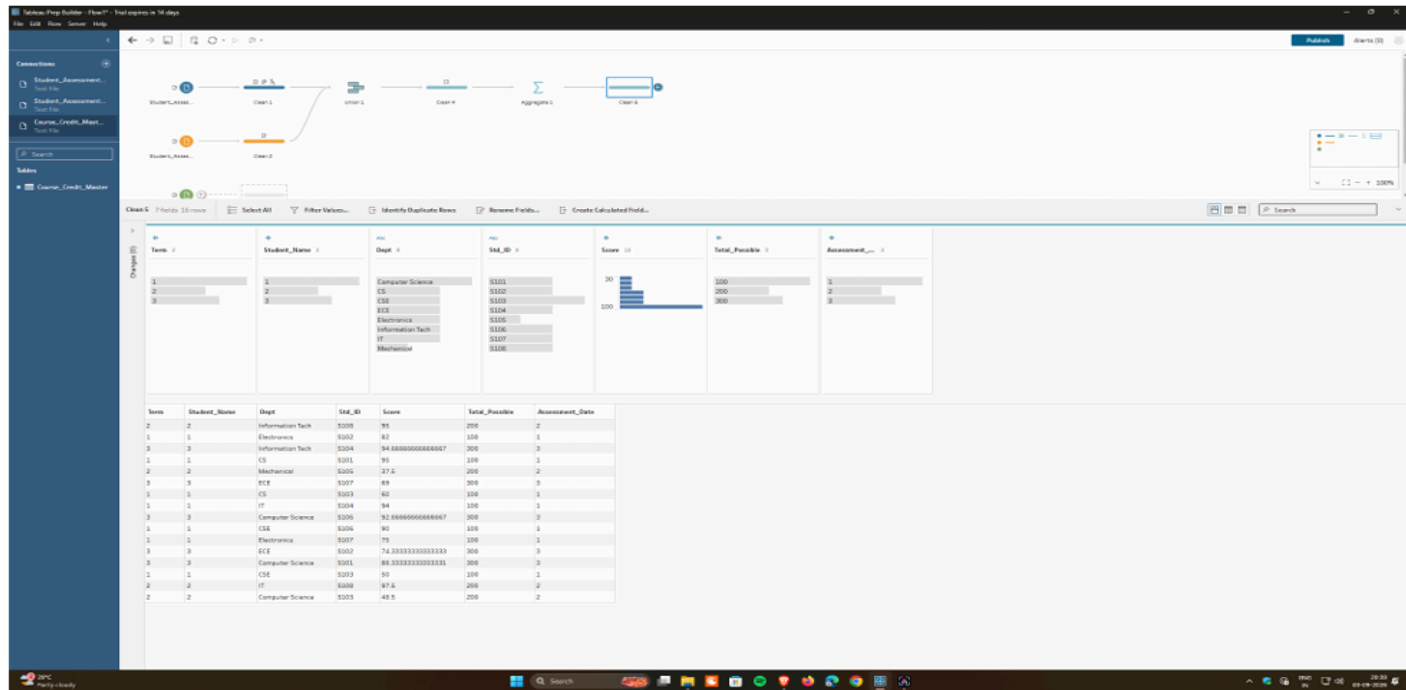
Look at the Profile Pane: o The row count collapses from 30 individual assessment rows into 15 summary records or more (one row per student per term).

F: Apply Filters and Transformations

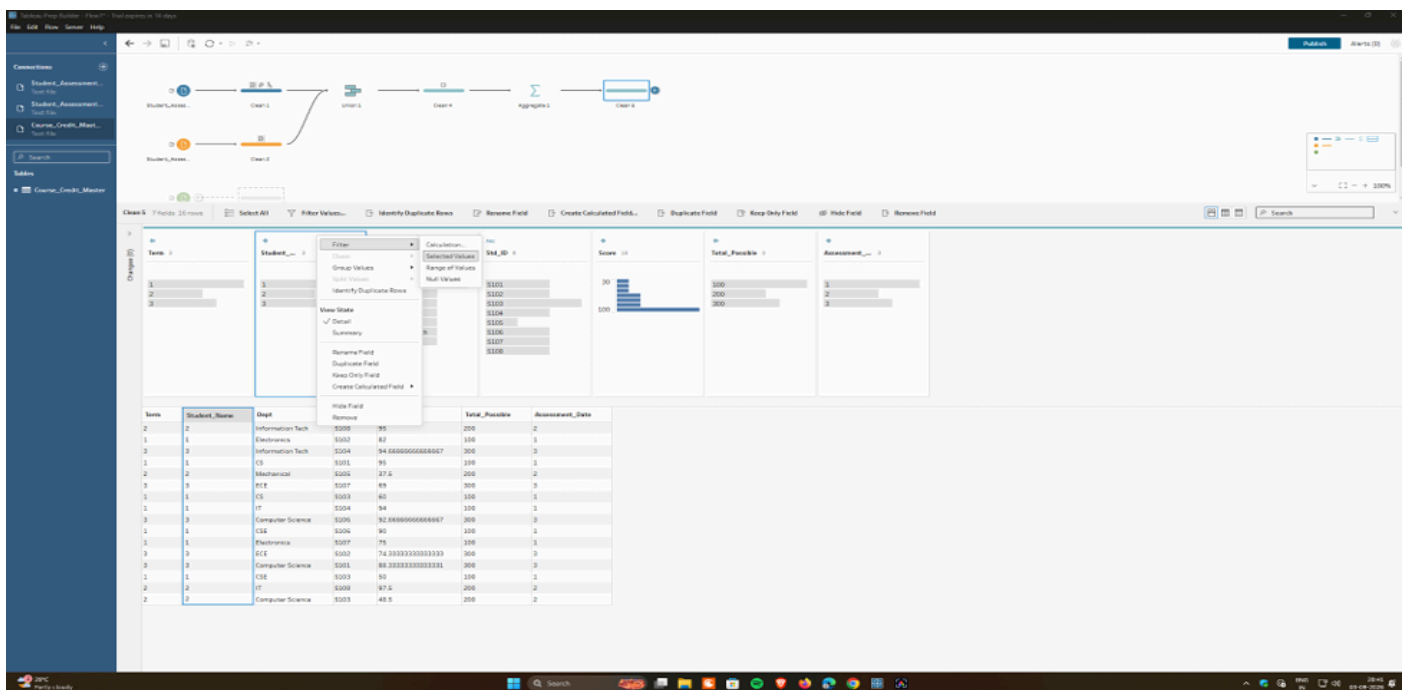
Step 1:- Click on the Clean 4 step connected after your Aggregate 1 step on the Flow canvas. The Profile Pane below will show your aggregated columns: Student\_ID, Student\_Name, Department, Term, Total\_Marks\_Obtained, Average\_Score, Total\_Max\_Marks, and Assessments\_Taken. (Naming Will be different and completely depends on how you performed the Union Step)

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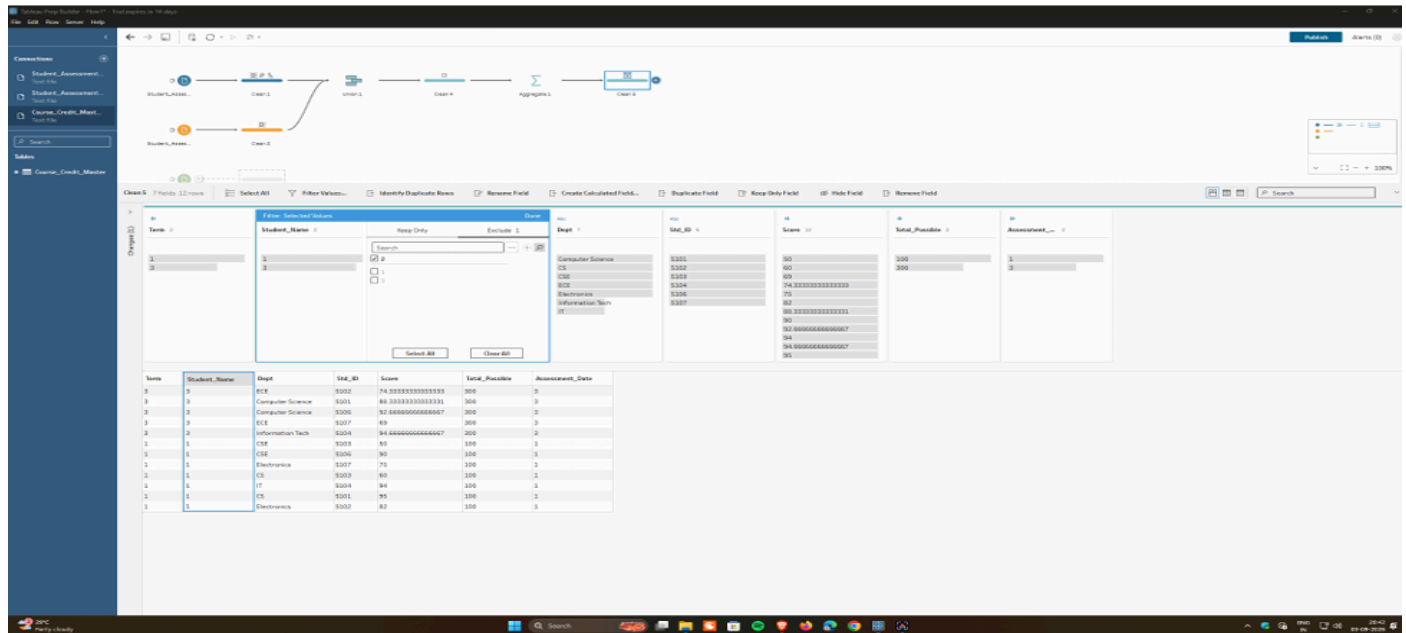
Step 2:- If Enrollment\_Status is present on a profile card: o Click on the card --> click the value Dropped --> right-click and select Exclude (or click the filter icon at the top of the card and check only Active). 2. (Alternative via Filter calculation): o In the top toolbar, click Filter Values --> Selected Values....



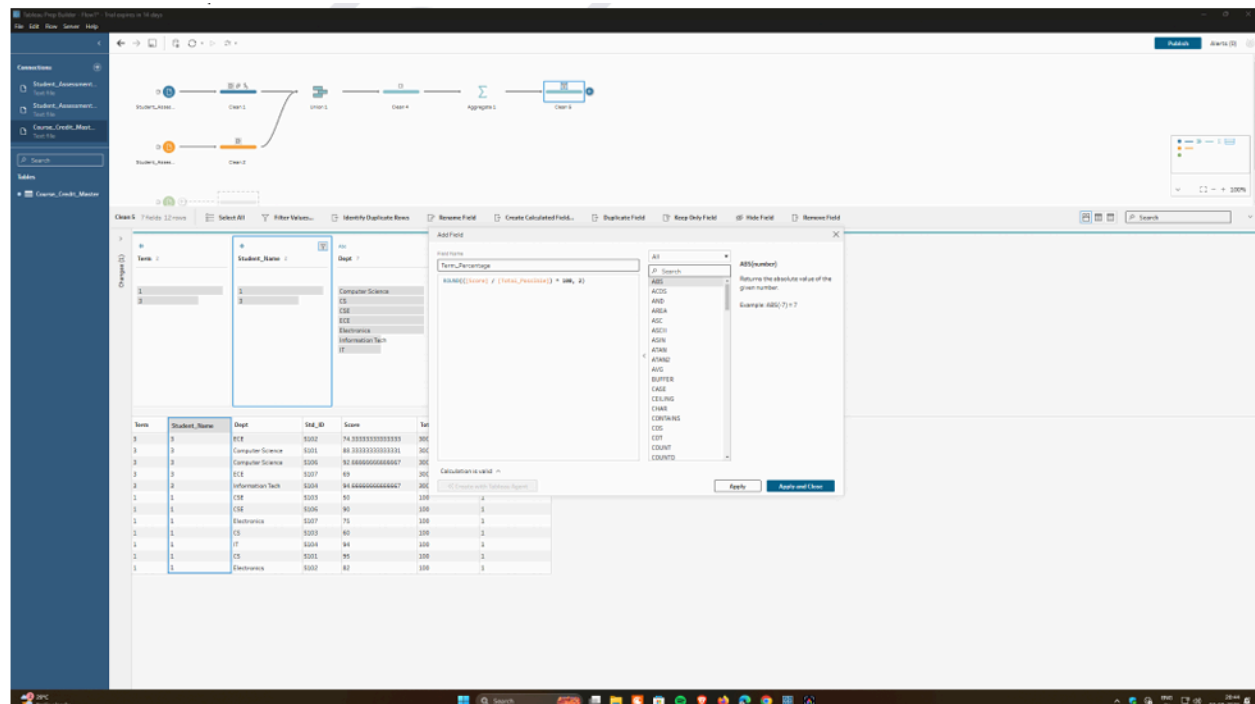
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Exclude Dropped so only Active students remain in the active pipeline.



Step 3:- Create the Normalized Term Percentage Metric 1. In the toolbar above the Profile cards, click Create Calculated Field. 2. Field Name: Term\_Percentage 3. Tableau Prep Formula: Code snippet  $\text{ROUND}([\text{Total\_Marks\_Obtained}] / [\text{Total\_Max\_Marks}] * 100, 2)$   $\text{ROUND}([\text{Score}] / [\text{Total\_Possible}] * 100, 2)$  {This formula can be used if you couldn't able to rename your aggregate fields.} 4. Click Save.

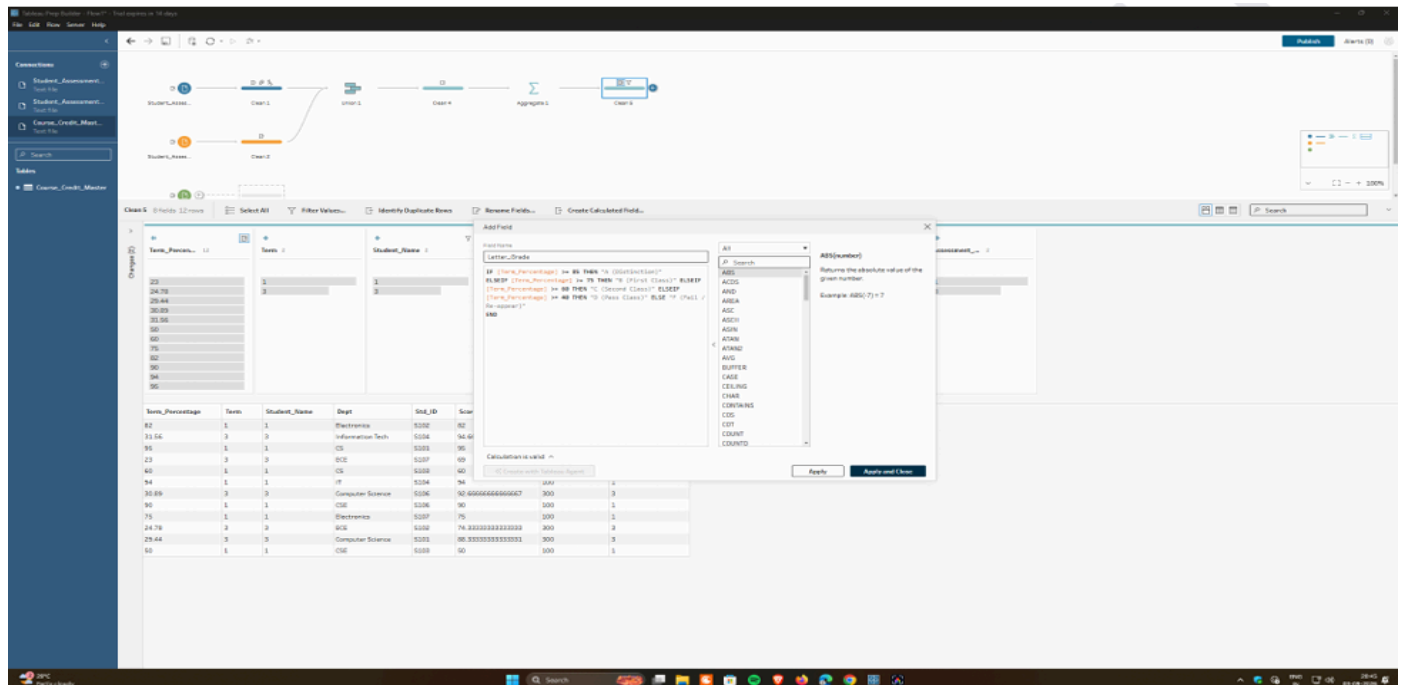


Step 4;- In the toolbar, click Create Calculated Field.

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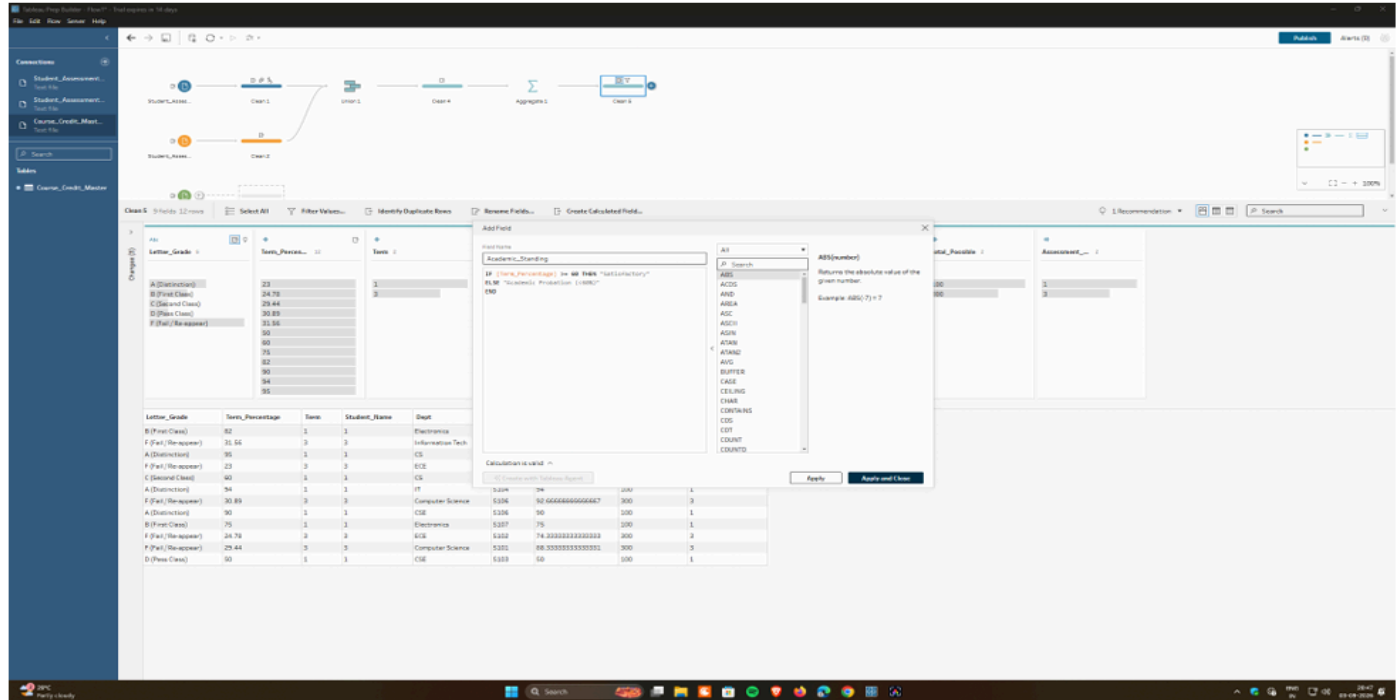
- Field Name: Letter\_Grade
- Tableau Prep Formula: Code snippet IF [Term\_Percentage] >= 85 THEN "A (Distinction)" ELSEIF [Term\_Percentage] >= 75 THEN "B (First Class)" ELSEIF [Term\_Percentage] >= 60 THEN "C (Second Class)" ELSEIF [Term\_Percentage] >= 40 THEN "D (Pass Class)" ELSE "F (Fail / Re-appear)" END
- Click Save. Step
- Create Academic Standing Compliance Flag :-
  - Click Create Calculated Field.
  - Field Name: Academic\_Standing
  - Tableau Prep Formula: Code snippet IF [Term\_Percentage] >= 60 THEN "Satisfactory" ELSE "Academic Probation (<60%)" END



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- Step 5:-
1. Click Create Calculated Field.
  2. Field Name: Academic\_Standing
  3. Tableau Prep Formula: Code snippet IF [Term\_Percentage] >= 60 THEN "Satisfactory" ELSE "Academic Probation (<60%)" END
  4. Click Save.



- Step 6:-
1. Look at the newly generated profile cards in Clean 4:
    - o Term\_Percentage: Displays a smooth numeric distribution between 0% and 100%.
    - o Letter\_Grade: Displays discrete categorical buckets (A (Distinction), B (First Class), etc.).
    - o Academic\_Standing: Highlights students requiring academic counseling. -

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G: Create and Automate Data Preparation Flows

Step 1:-

1. On the Flow canvas, locate the final cleaning step (Clean 4).
2. Hover over the Clean 4 icon and click the + icon.
3. Select Output (represented by the download/save arrow icon).
4. An Output step node will appear connected directly to Clean 4

| Letter_Grade         | Term_Percentage | Academic_Standings        | Score | Student_Name | Dept             | Std_ID | Score | Total_Possible | Assessment_Date |
|----------------------|-----------------|---------------------------|-------|--------------|------------------|--------|-------|----------------|-----------------|
| A (Distinction)      | 23              | Academic Probation (<60%) | 1     | 1            | Electronics      | S102   | 62    | 100            | 1               |
| B (First Class)      | 24.75           | Satisfactory              | 3     | 3            | Information Tech | S104   | 94    | 100            | 3               |
| C (Second Class)     | 29.44           | Satisfactory              | 1     | 1            | CS               | S103   | 95    | 100            | 1               |
| D (Pass Class)       | 30.89           | Academic Probation (<60%) | 3     | 3            | IT               | S107   | 69    | 100            | 3               |
| F (Fail / Re-assess) | 31.56           | Satisfactory              | 1     | 1            | CS               | S103   | 60    | 100            | 1               |
|                      | 95              |                           |       |              | IT               | S104   | 94    | 100            | 1               |
|                      | 95              |                           |       |              | Computer Science | S106   | 90    | 100            | 1               |
|                      | 95              |                           |       |              | Electronics      | S107   | 75    | 100            | 1               |
|                      | 95              |                           |       |              | Information Tech | S103   | 95    | 100            | 1               |
|                      | 95              |                           |       |              | IT               | S104   | 94    | 100            | 1               |

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Step 2:- Click on the Output node to open its configuration panel below:

- o Save output to: Select File (or Tableau Server / Tableau Cloud).
- o Output type: Select Tableau Data Extract (.hyper) for high-speed dashboarding, or Comma Delimited Text (.csv) for universal export.
- o Location: Click Browse and choose the directory where you want the final dataset saved.
- o File Name: Enter Student\_Performance\_Master\_Cleaned.
- o Write Option: Select Create table (or Append to table if updating existing records).

The screenshot displays the Power BI Desktop interface. On the left, the 'Connections' pane shows data sources: 'Student\_Assessment', 'Student\_Assessment...', 'Course\_Credit\_Mast...', and 'Course\_Credit\_Mast...'. The main workspace shows a data flow diagram with nodes: 'Student\_Assessment', 'Student\_Assessment...', 'Course\_Credit\_Mast...', 'Course\_Credit\_Mast...', 'Aggregates', 'Course\_Credit\_Mast...', and 'Output'. The 'Output' node is selected, and its configuration panel is open, highlighted with a red box. The configuration panel includes the following options:

- Save output to:** File (selected)
- Output type:** Tableau Data Extract (.hyper)
- Location:** Browse
- File name:** Student\_Performance\_Master\_Cleaned
- Write options:** Create table (selected)

The background of the screenshot shows a data table with columns: 'Student\_Grade', 'Term\_Percentage', 'Academic\_Standings', 'Term', 'Student\_Name', 'Dept', 'Std\_ID', 'Score', 'Total\_Possible', and 'Assessment\_Date'. The table contains 10 rows of data.

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- Step 3:- 1. In the Output pane at the bottom-left, click the blue Run Flow button (or click the play icon  $\blacktriangleright$  directly on the Output node on the canvas).
2. Tableau Prep will process all upstream nodes (Cleaning  $\rightarrow$  Grouping  $\rightarrow$  Unioning  $\rightarrow$  Merging  $\rightarrow$  Aggregating  $\rightarrow$  Scoring).
3. A success notification will appear stating: "The flow ran successfully in X seconds."

The screenshot displays the Tableau Prep Studio interface. The top canvas shows a data flow with nodes: 'Student\_Assess', 'Clean 1', 'Clean 2', 'Clean 3', 'Aggregate 1', 'Clean 4', and 'Score'. The bottom pane shows the 'Output' section with a table titled 'Student\_Performance\_Master\_Cleaned\_Score'. A modal window titled 'Finished Running Flow' is overlaid on the table, indicating the flow completed successfully in 00:00:00. The table contains columns: Letter\_Grade, Score\_Percentage, Academic\_Standing, Score, Student\_Name, Dept, Std\_ID, Score, Total\_Possible, and Assessment\_Date.

| Letter_Grade         | Score_Percentage | Academic_Standing         | Score | Student_Name | Dept             | Std_ID | Score             | Total_Possible | Assessment_Date |
|----------------------|------------------|---------------------------|-------|--------------|------------------|--------|-------------------|----------------|-----------------|
| B (First Class)      | 82               | Satisfactory              | 3     | 1            | Electronics      | 0002   | 82                | 100            | 3               |
| F (Fail / No appear) | 25.25            | Academic Probation (<40%) | 3     | 3            | Information Tech | 0004   | 54.99999999999999 | 200            | 3               |
| A (Distinction)      | 94               | Satisfactory              | 3     | 1            |                  |        |                   | 100            | 3               |
| F (Fail / No appear) | 23               | Academic Probation (<40%) | 3     | 3            |                  |        |                   | 300            | 3               |
| C (Second Class)     | 60               | Satisfactory              | 3     | 1            |                  |        |                   | 100            | 3               |
| A (Distinction)      | 94               | Satisfactory              | 3     | 1            |                  |        |                   | 100            | 3               |
| F (Fail / No appear) | 30.89            | Academic Probation (<40%) | 3     | 3            |                  |        | 6666666666666667  | 300            | 3               |
| A (Distinction)      | 94               | Satisfactory              | 3     | 1            |                  |        |                   | 100            | 3               |
| B (First Class)      | 79               | Satisfactory              | 3     | 1            |                  |        |                   | 100            | 3               |
| F (Fail / No appear) | 24.75            | Academic Probation (<40%) | 3     | 3            |                  |        | 0000000000000000  | 300            | 3               |
| F (Fail / No appear) | 29.44            | Academic Probation (<40%) | 3     | 3            |                  |        | 0000000000000000  | 300            | 3               |
| D (Pass Class)       | 50               | Academic Probation (<40%) | 3     | 1            |                  |        |                   | 100            | 3               |

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The screenshot displays a Windows desktop environment. On the left, a File Explorer window is open to the 'Desktop' location, showing a folder named 'Student\_Performance\_Paper\_10'. The 'File name' field contains 'Student\_Performance\_Paper\_10', and the 'Save as type' is set to 'Microsoft PowerPoint 97-2003'. Below the file explorer, a presentation slide is visible, titled 'Winits Options'. The slide content includes a table with columns for 'Winits Options', 'Academic Production (%)', and 'Academic Production (%)'. The table has two rows of data. Below the table, there is a section for 'Full ratings' and a 'Create table' button.

On the right side of the desktop, a presentation slide is displayed. The slide title is 'Winits Options'. The slide content includes a table with columns for 'Winits Options', 'Academic Production (%)', and 'Academic Production (%)'. The table has two rows of data. Below the table, there is a section for 'Full ratings' and a 'Create table' button.

Below the presentation slide, a data table is visible. The table has columns for 'Dept', 'Win\_ID', 'Score', 'Total\_Provide', and 'Assessment\_Date'. The table contains 15 rows of data.

| Dept             | Win_ID | Score | Total_Provide | Assessment_Date |
|------------------|--------|-------|---------------|-----------------|
| Electronics      | 10002  | 90    | 100           | 1               |
| Information Tech | 10004  | 94    | 100           | 2               |
| IS               | 10005  | 94    | 100           | 1               |
| IT               | 10007  | 89    | 100           | 3               |
| IT               | 10009  | 40    | 100           | 1               |
| IT               | 10004  | 94    | 100           | 1               |
| Computer Science | 10006  | 92    | 100           | 2               |
| CS               | 10004  | 90    | 100           | 1               |
| Electronics      | 10007  | 75    | 100           | 2               |
| ECG              | 10002  | 74    | 100           | 3               |
| Computer Science | 10005  | 96    | 100           | 1               |
| CSE              | 10009  | 90    | 100           | 1               |

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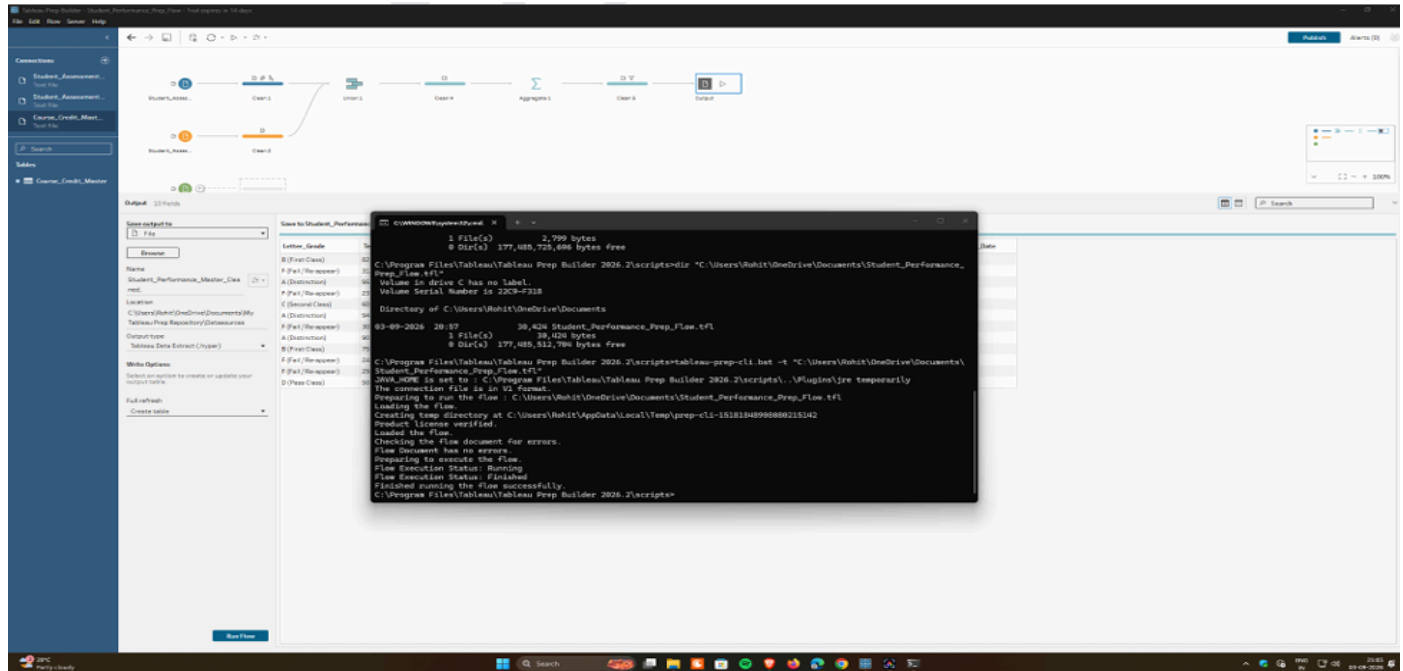
## SUBJECT NAME: Power BI And Tableau Practical

Step 5:- (Automation Options in Production):

o Command Line Execution: You can automate this flow on Windows/Mac using the Tableau Prep Command Line Interface (CLI): `Bash tableau-prep-cli -t`

`Student_Performance_Prep_Flow.tfl`

o Tableau Prep Conductor: If connected to Tableau Cloud/Server, you can publish the flow and set a scheduled refresh (e.g., every Monday at 6:00 AM) to automatically run whenever new semester files are added.

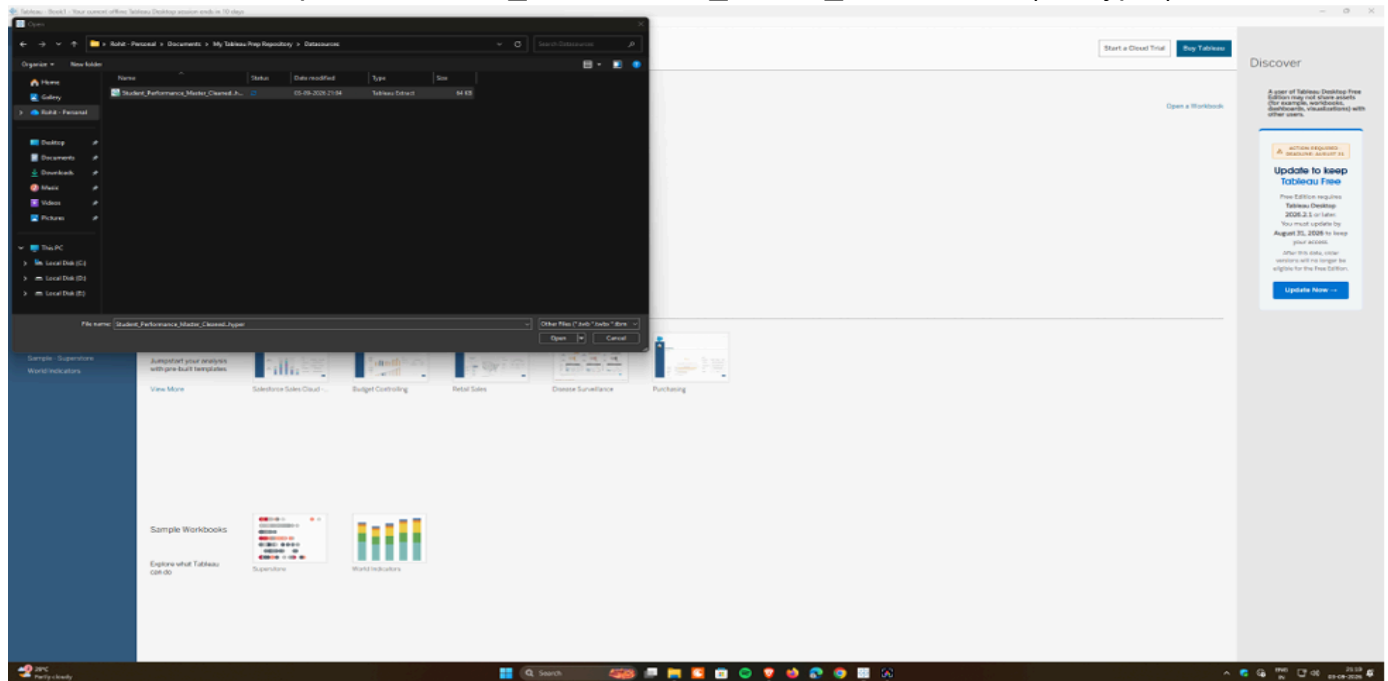


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Step 6:- Verify the Final Master Dataset in Tableau Desktop

1. Open Tableau Desktop.
2. Connect to the exported Student\_Performance\_Master\_Cleaned.csv (or .hyper).



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Build the self-performance dashboard using the pre-calculated metrics:

- o Drag Student\_Name to Rows.
- o Drag Term to Columns.
- o Drag Term\_Percentage to Text.
- o Drag Letter\_Grade to Color.
- o Drag Academic\_Standing to Tooltip.

